Histology Lecture

MANIPAL
UNIVERSITY COLLEGE
MALAYSIA

Formerly known as MELAKA-MANIPAL MEDICAL COLLEGE

Small Intestine

Dr. Lalu Krishna



#inspire

Learning Objectives

At the end of the lecture, students should be able to:

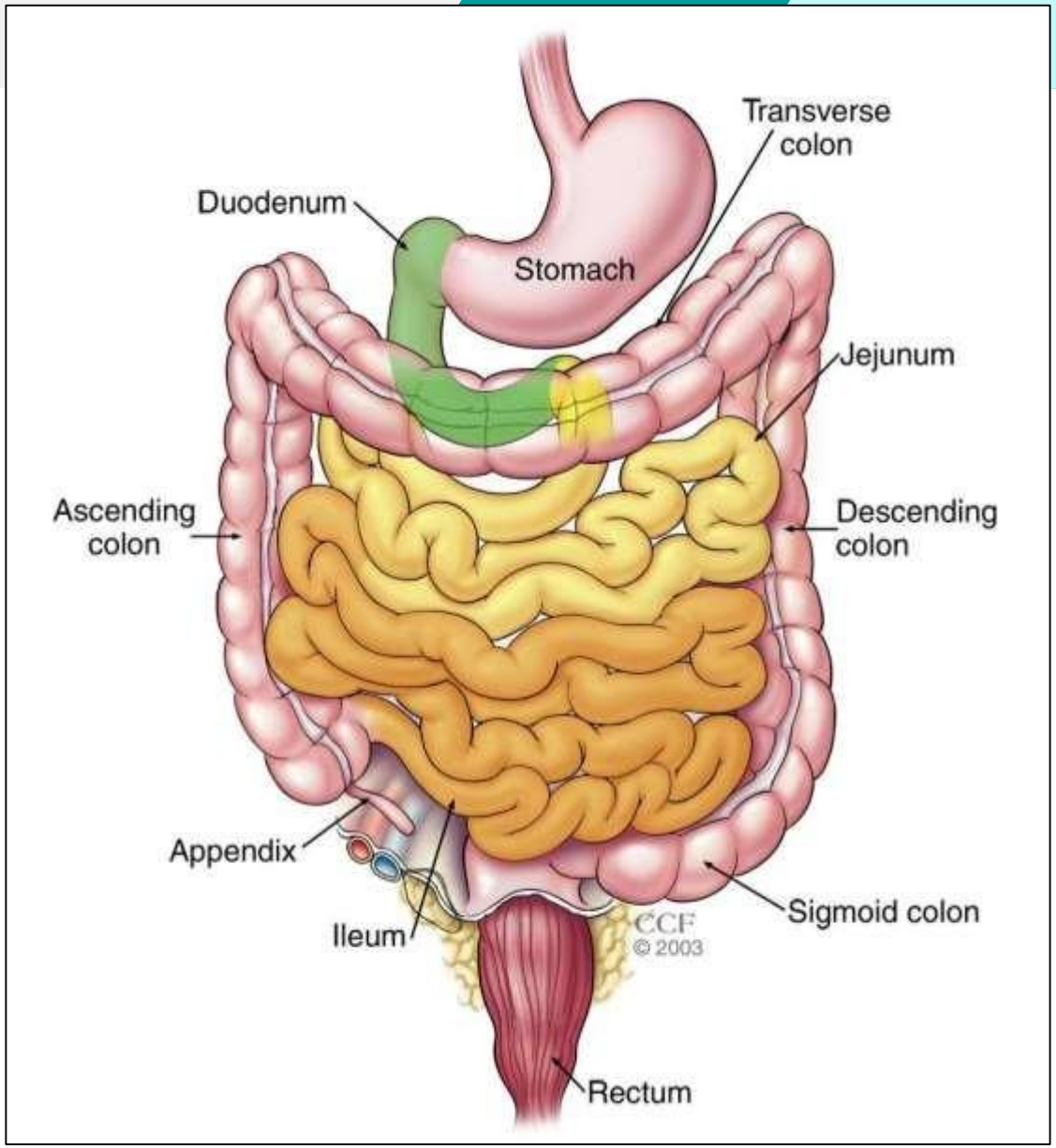
- Small intestine, large Intestine, appendix
  - Explain and illustrate the microscopic structure of small intestine
  - Describe the different parts of small intestine

Small intestine

- Muscular tube which extend from the pyloric sphincter to the ileocecal valve. It is suspended from posterior abdominal wall by the mesentery.

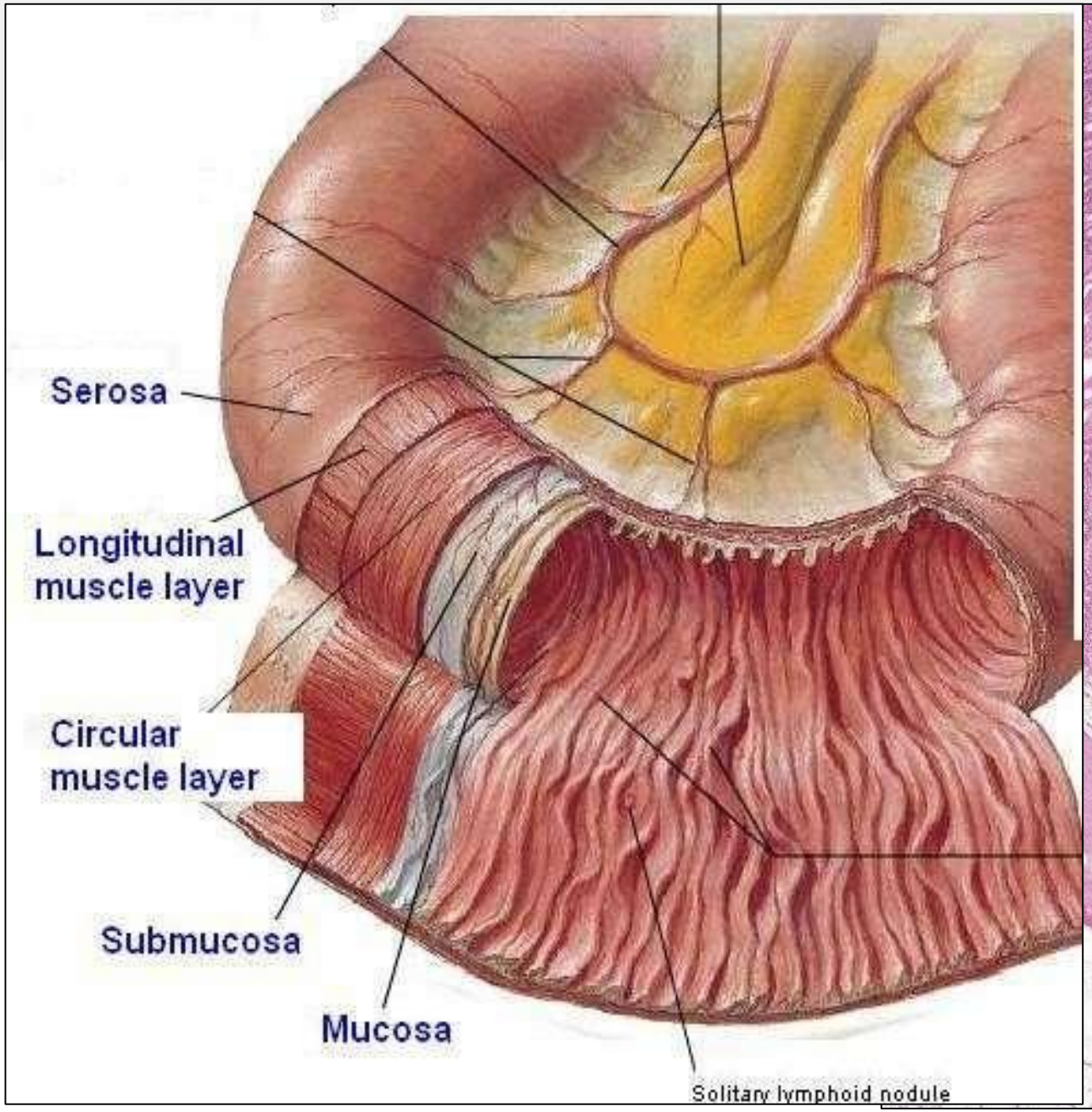
- Major site of nutrient absorption into the blood stream.

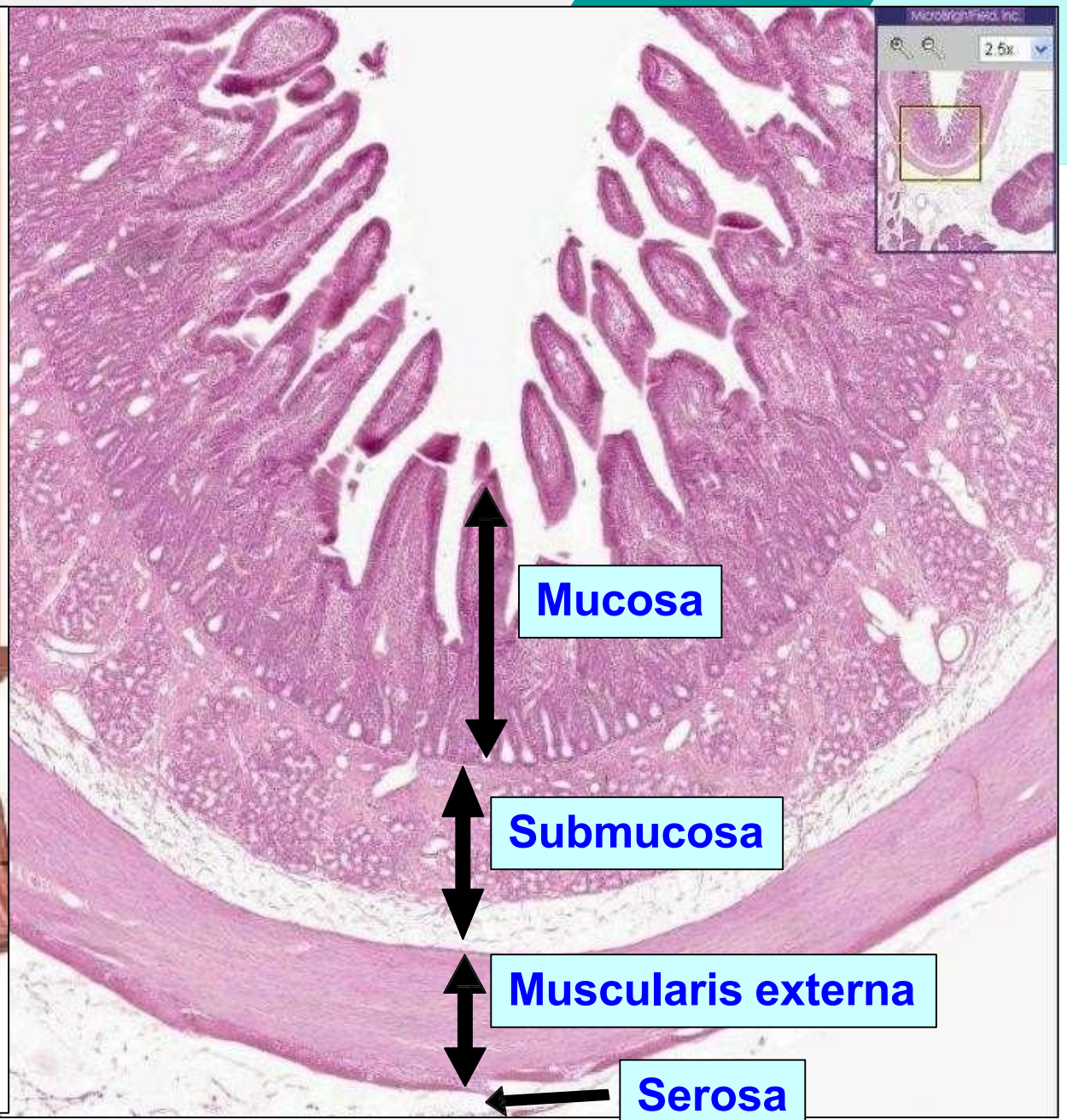
- Small intestine is subdivided into:
  - ✓**Duodenum**
  - ✓**Jejunum**
  - ✓**Ileum**



General histology

- Histologically, it has four distinctive layers.
  1. **Mucosa:**
     - Epithelium
     - Lamina propria
     - Muscularis mucosae
  2. **Submucosa**
  3. **Muscularis externa:**
     - Inner circular
     - Outer longitudinal
  4. **Serosa/ Adventitia**


Serosa
Longitudinal muscle layer
Circular muscle layer
Submucosa
Mucosa
Solitary lymphoid nodule

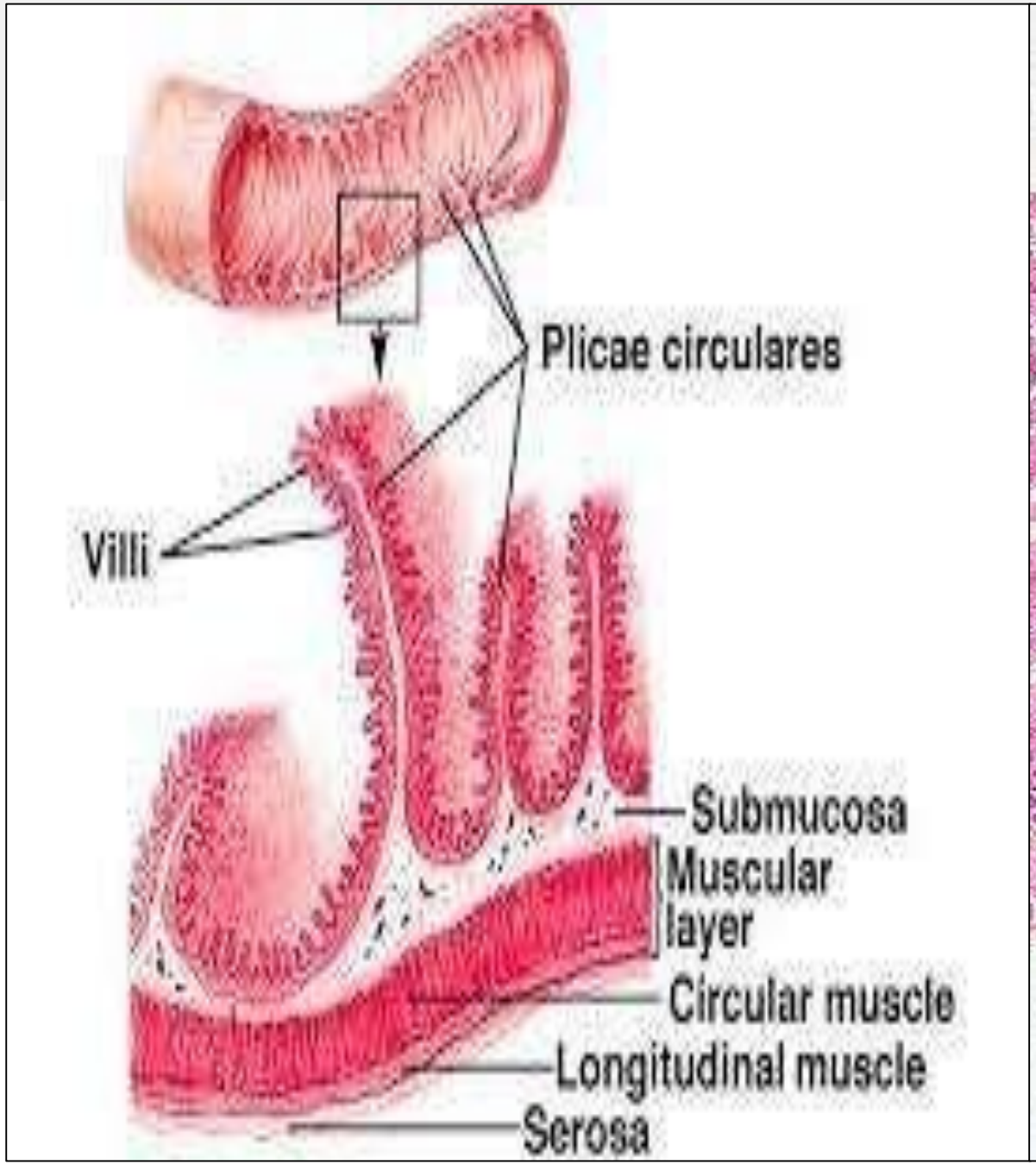

Mucosa
Submucosa
Muscularis externa
Serosa

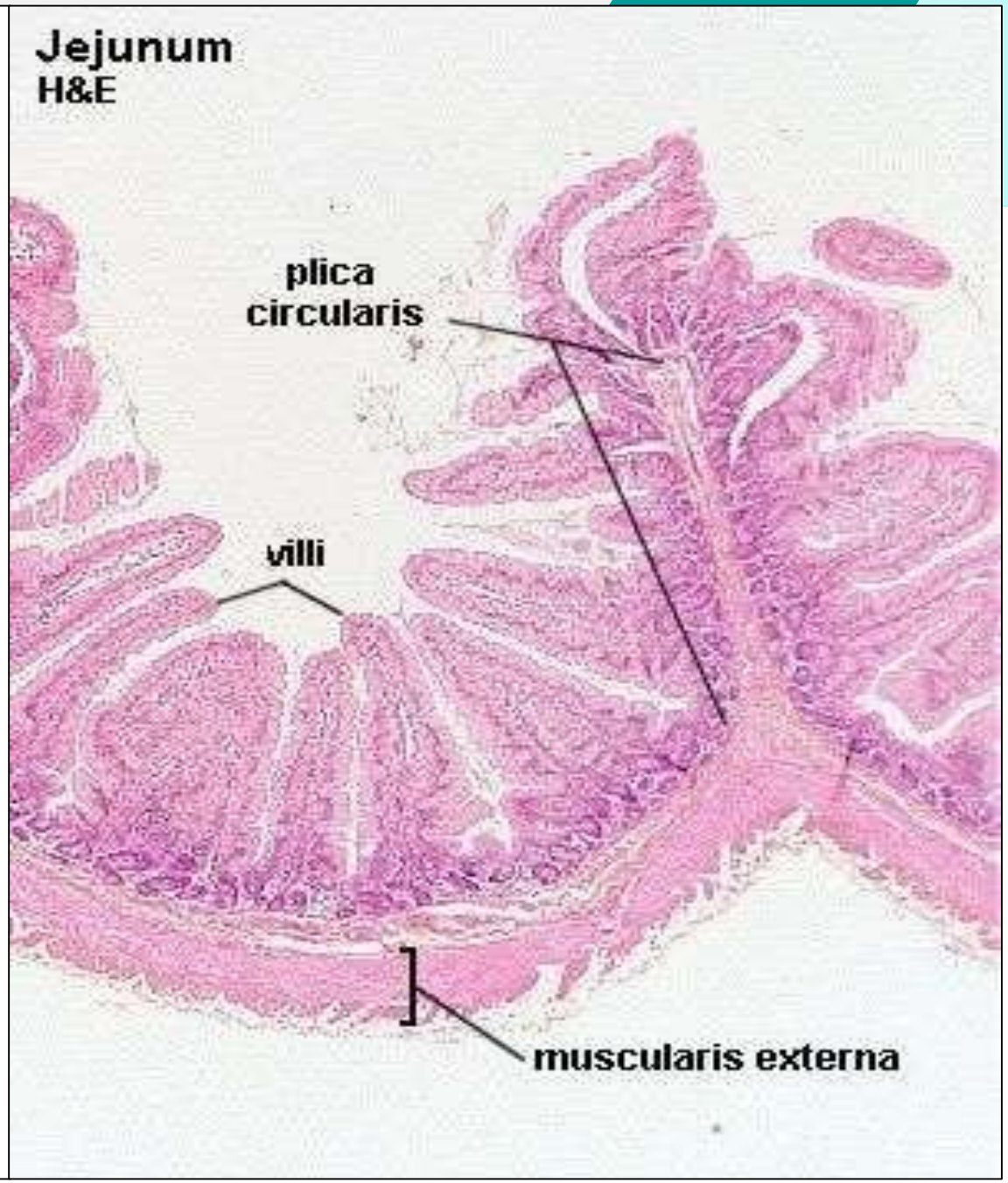
Modifications of internal surface of small intestine

- Internal surface of small intestine is modified to increase its surface area by:

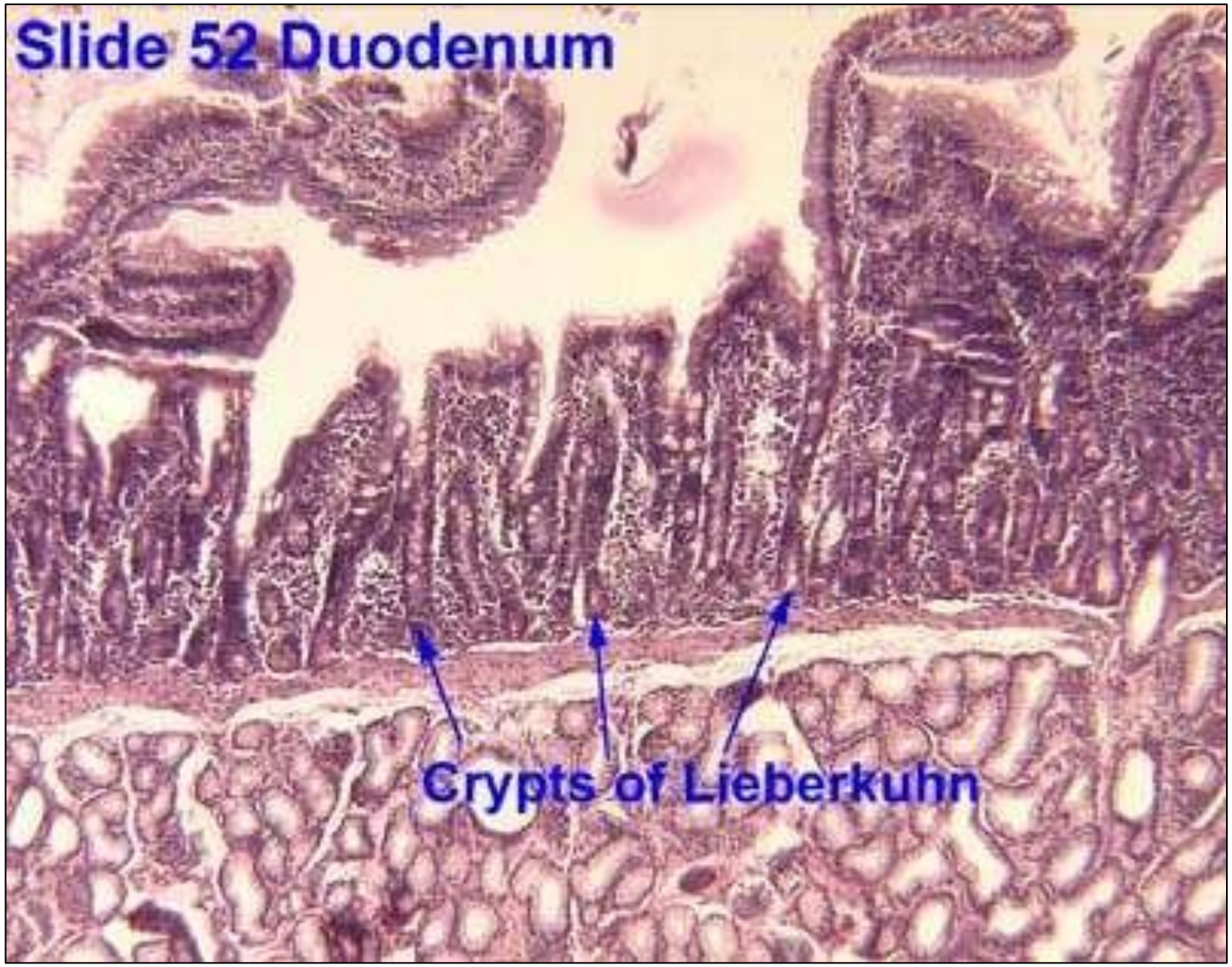
1) **Plicae circulares** (circular fold/valves of Kerckring): Spiral fold of mucosa and submucosa which help to increase the surface area by 2-3 folds

2) **Intestinal villi:** Finger-like projection of mucosa which help to increase the surface area by 10-folds


Plicae circulares
Villi
Submucosa
Muscular layer
Circular muscle
Longitudinal muscle
Serosa


Jejunum
H&E
plica circularis
villi
muscularis externa

- Epithelium of small intestine invaginates between villi to form **crypts of Lieberkuhn** (intestinal gland).



Mucosa

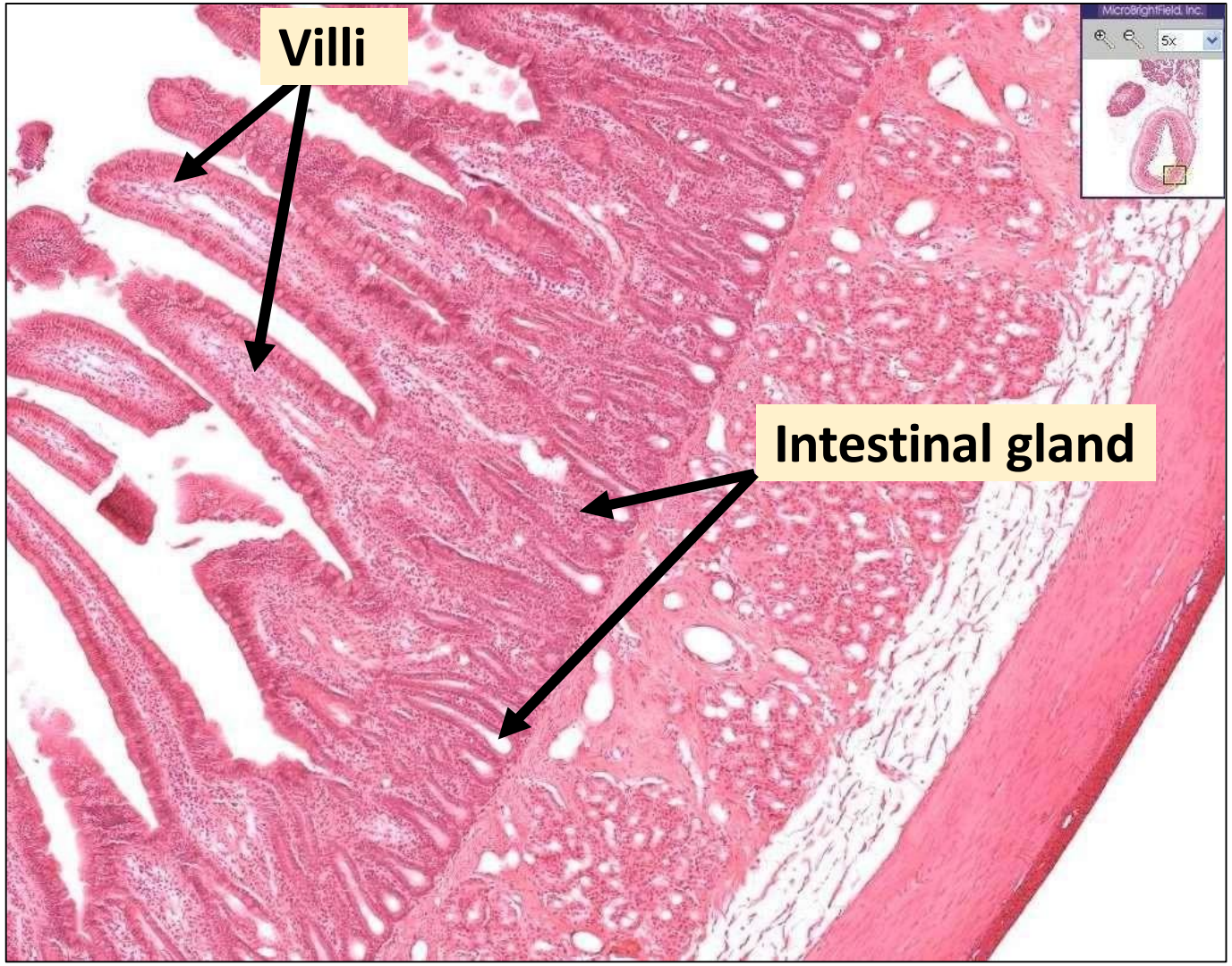
1) **Epithelium**
   - Simple columnar epithelium
   - Is composed of absorptive cells, goblet cells and some enteroendocrine cells

2) **Lamina propria**
   - Occupies the cores of villi and spaces between crypt of Lieberkuhn
   - Contains crypt of Lieberkuhn, lymphoid follicles, blood vessels, nerves

3) **Muscularis mucosa**
   - Smooth muscle cells that produce local movements of mucosa
   - Two layer - inner circular and outer longitudinal layer

Crypt of Lieberkuhn

- Simple tubular gland
- Opens into intervillous spaces
- Contains absorptive cells, goblet cells, stem cells, Paneth cells, and enteroendocrine cells.
  - ✓ Absorptive & goblet cells occupies mostly the upper half of the gland
  - ✓ Stem cells, Paneth cells & enteroendocrine cells occupies lower half of the gland



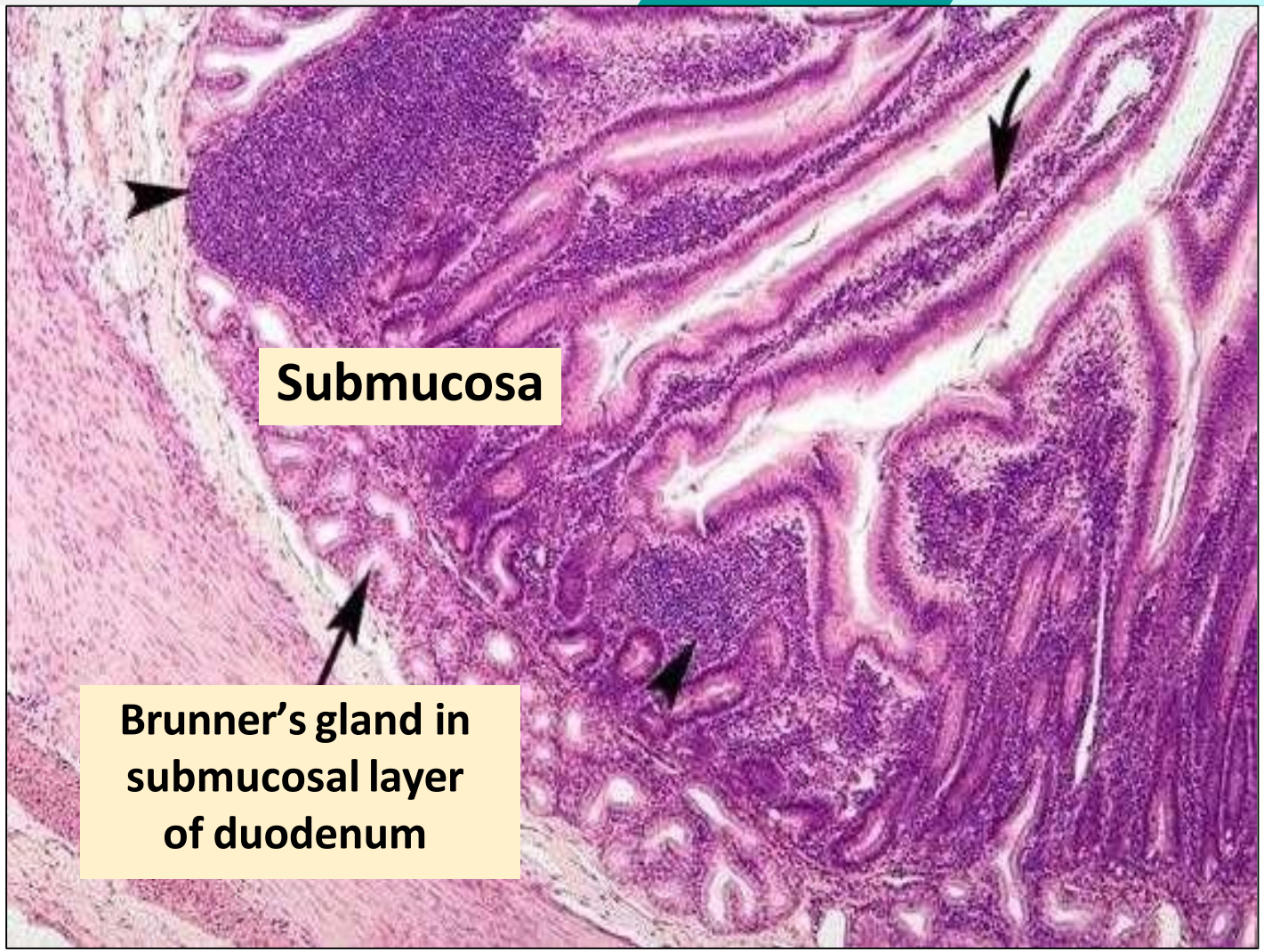
Submucosa

- Composed of **dense irregular connective tissue** containing blood vessels, nerve fibers and submucosal plexus (Meissner's plexus).

- In duodenum, **Brunner's glands** are also present in submucosa
  - ✓ Help to distinguish duodenum from jejunum and ileum

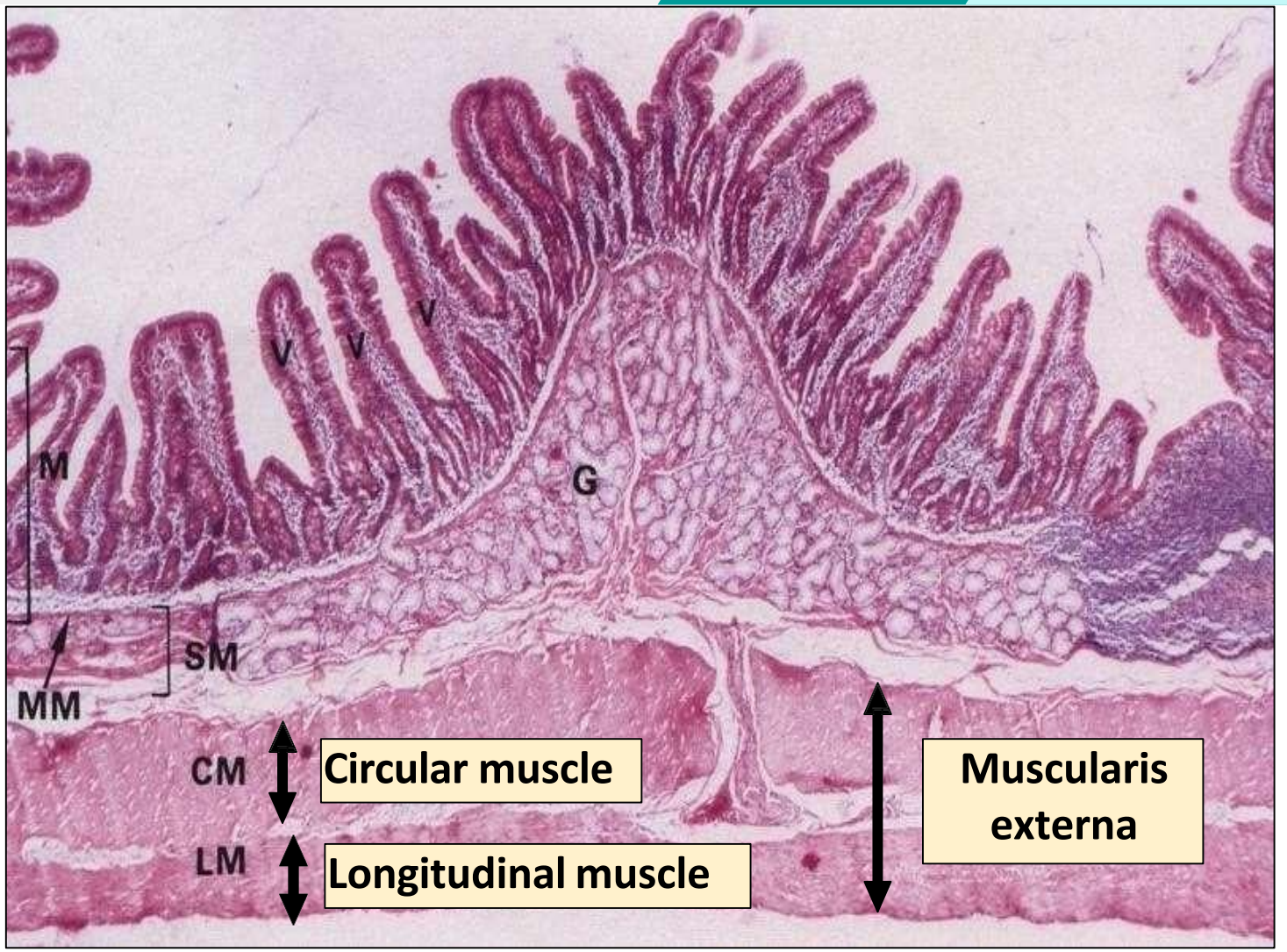
Brunner’s glands

- Only present in duodenum.
- Branched tubuloacinar glands in submucosa layer.
- Their ducts usually opens into the bases of crypt of Lieberkuhn.
- Secrete mucous alkaline fluid (to neutralize acid)



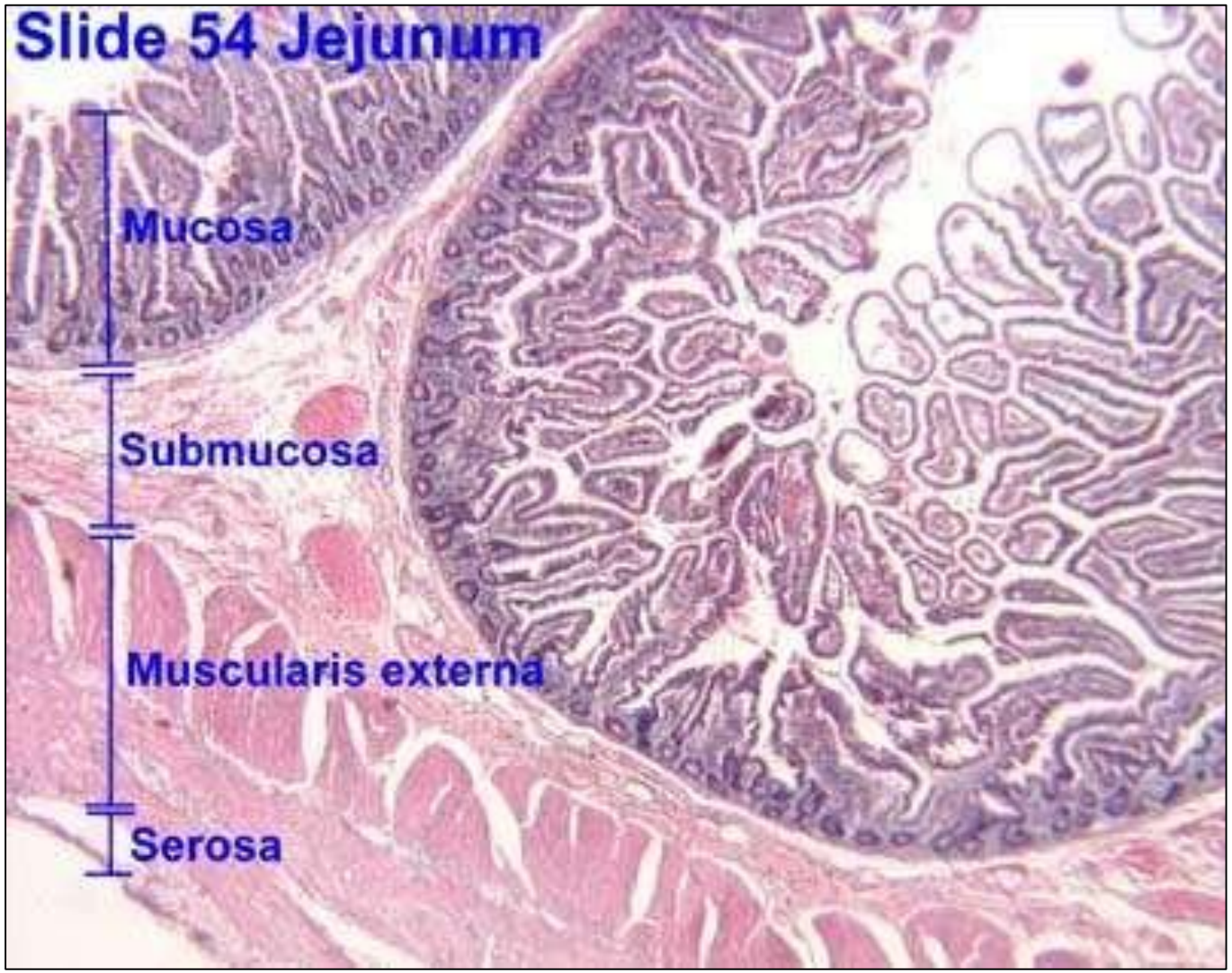
Muscularis externa

- Composed of **inner circular** layer and **outer longitudinal** smooth muscle layer.
- **Myenteric (Auerbach's) plexus** is located between these 2 muscle layers.
- Responsible for segmentation and peristalsis of small intestine.



Serosa/ Adventitia

- **Serosa** = From visceral peritoneum
- Serosa covers **all of the jejunum, ileum and first part of duodenum**.
- **Adventitia** covers the **remainder of the duodenum**.



Cells of the small intestine

1) Absorptive cells (Enterocytes)

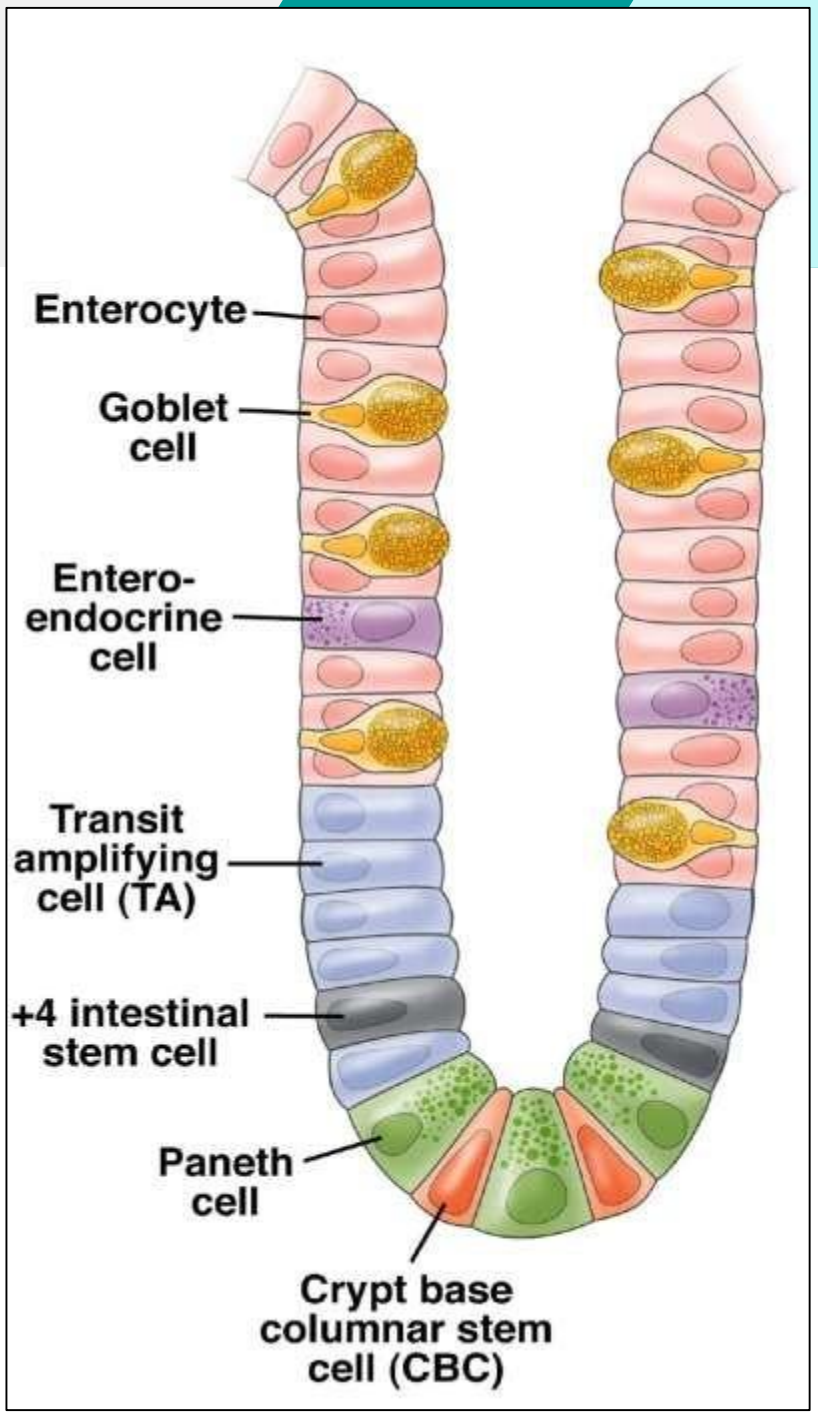
- Tall columnar cells with basally located nuclei
- Possess microvilli (brush border) on apical surface
- Have a glycocalyx which binds various enzymes
- Function: Terminal digestion of dipeptides & disacharides and absorption of nutrients

2) Goblet cells

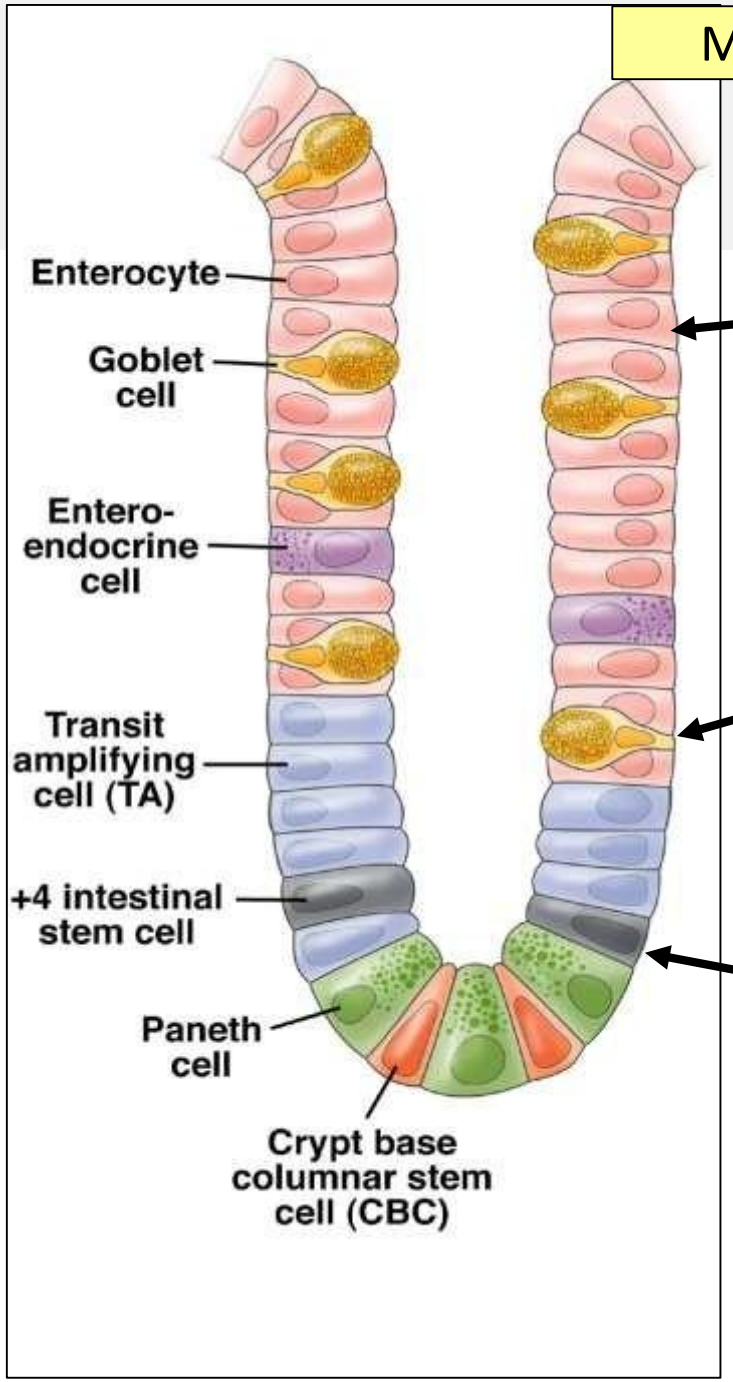
- Unicellular gland that produce mucinogen
- The number of goblet cells increase from duodenum to ileum

3) Stem cells

- Located in the basal half of the crypt of Lieberkuhn
- Function: To replace epithelium of the crypt, mucosal surface and villus



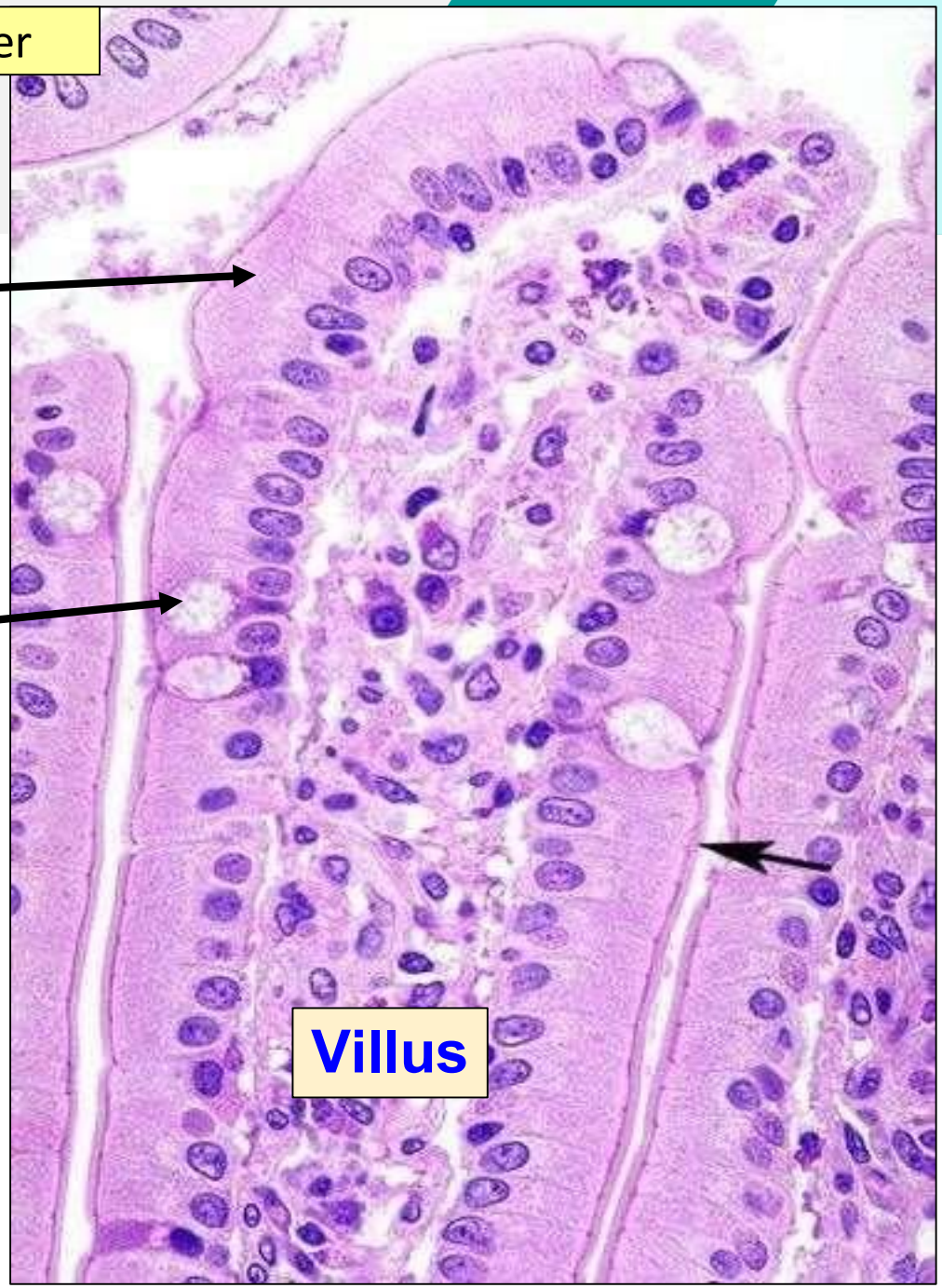
Microvilli forming brush border



Enterocytes

Goblet cell

Stem cell



4) Paneth cells

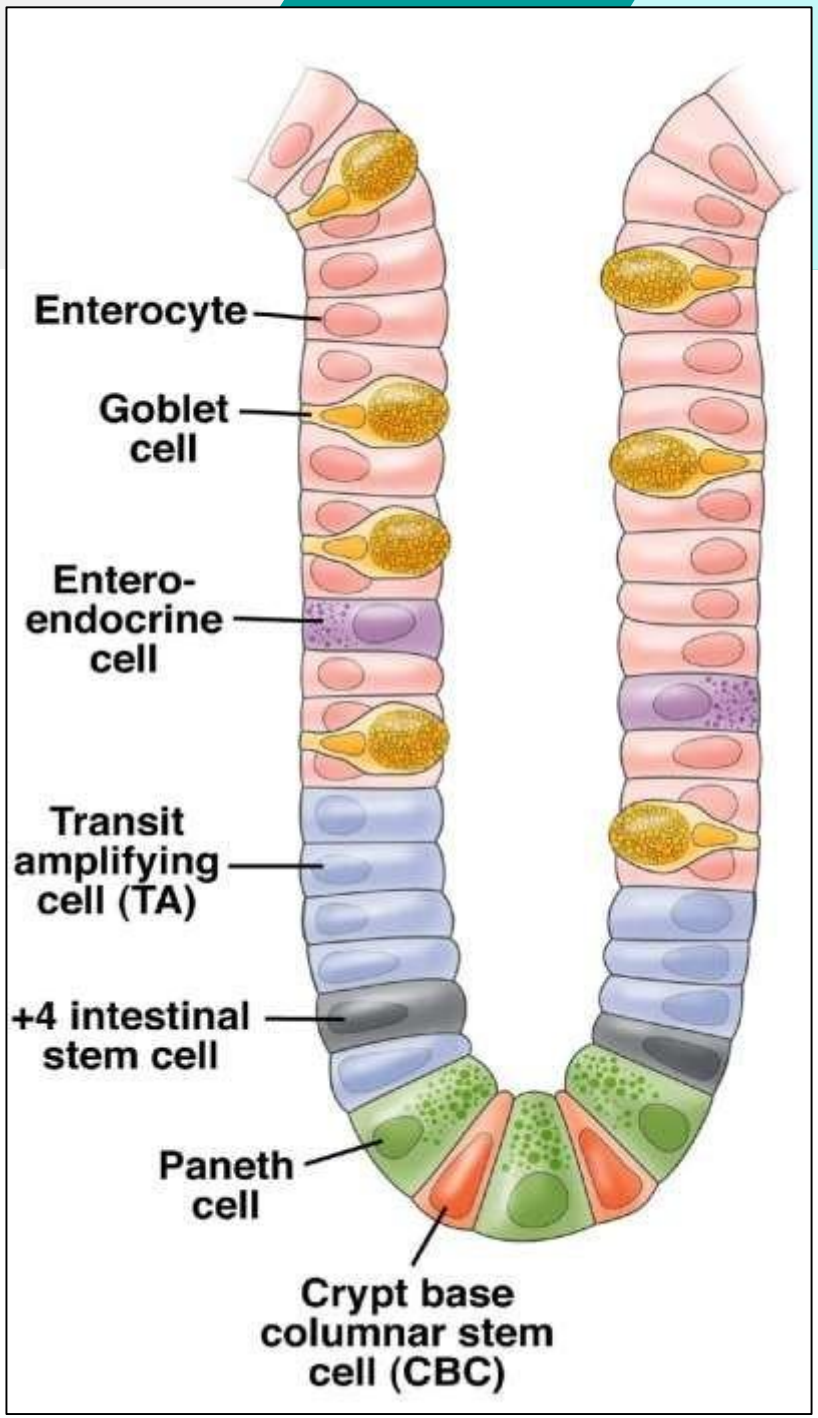
- Pyramidal-shaped cells with apical granules
- Located at the base of the crypt of Lieberkuhn
- Function: protects intestine from invading bacteria & secrete antibacterial enzyme lysozyme

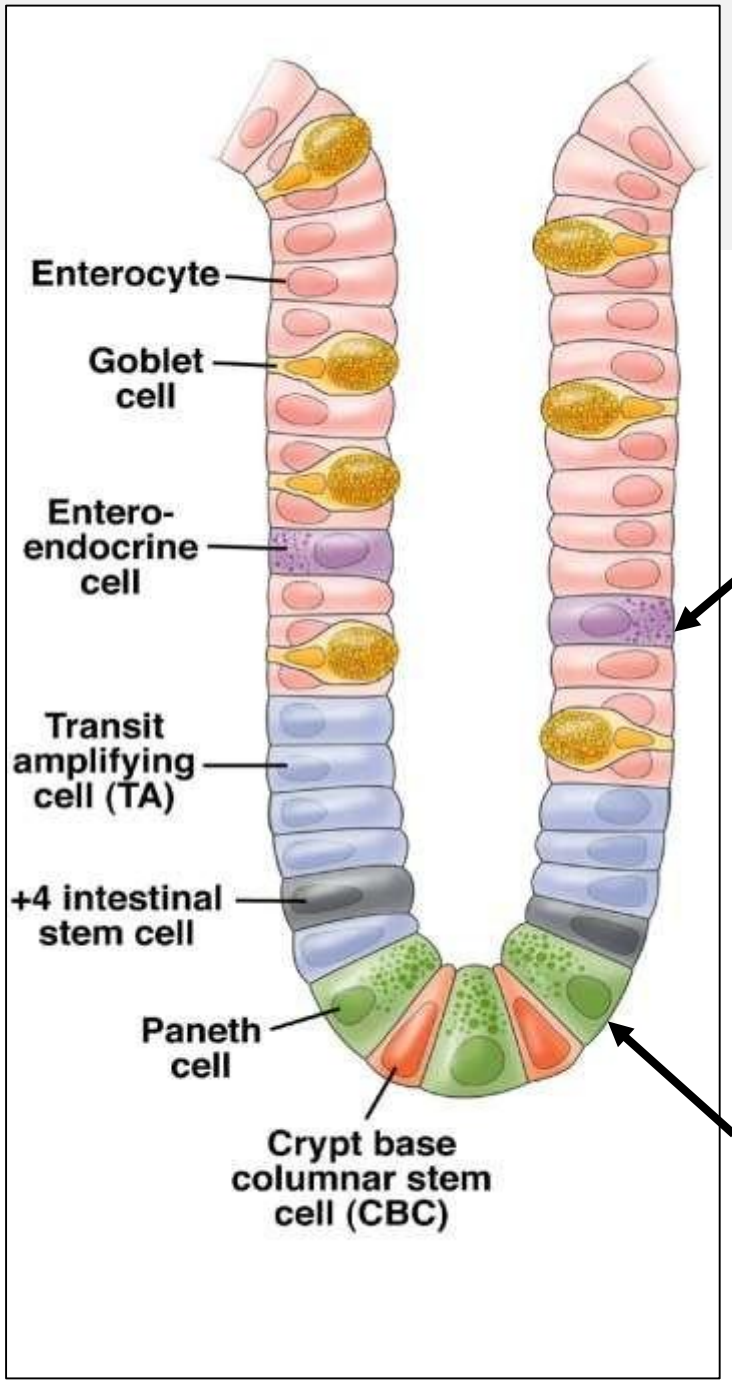
5) Enteroendocrine cells

- Granules are basally located (because hormones are secreted into surrounding blood vessels)
- Function: secrete various hormones such as gastrin, cholecytokinin, gastric inhibitory peptide

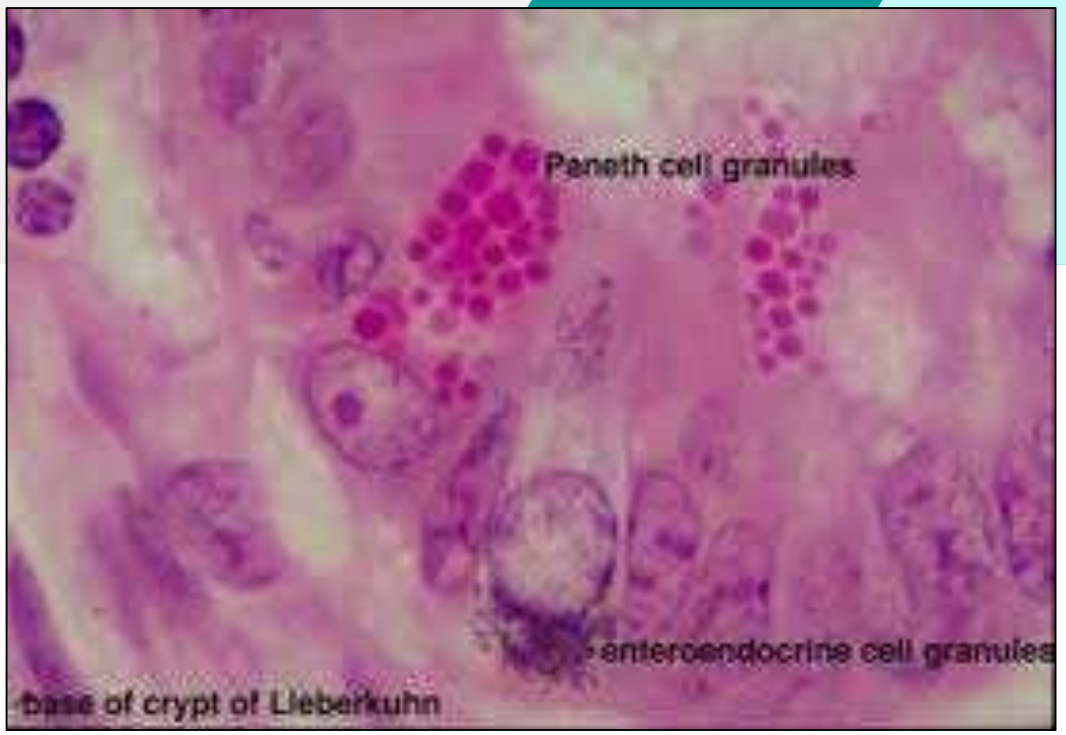
6) M cells (microfold cells)

- Belong to mononuclear phagocytic system
- Lie in the epithelium over lymphoid nodules
- Function: phagocytose antigen in intestinal lumen and transport them to macrophages & lymphocytes in lamina propria





Enteroendocrine cell



M-cells

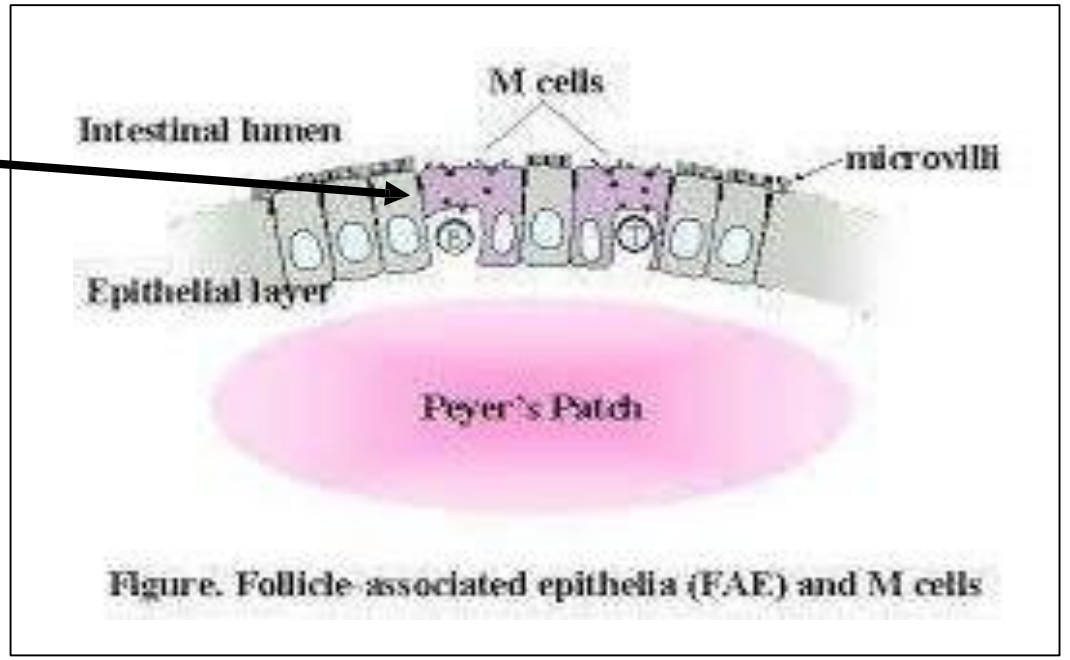


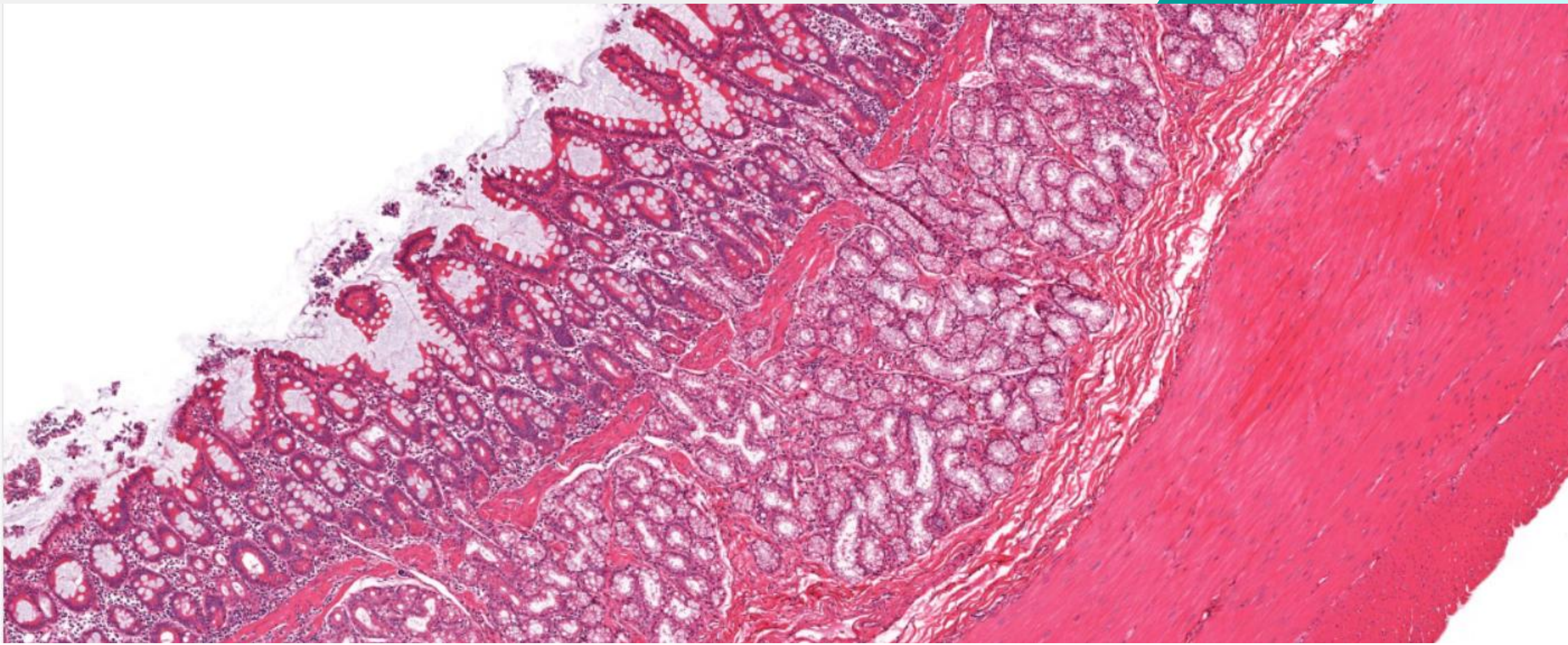
Figure. Follicle associated epithelia (FAE) and M cells

Paneth cells

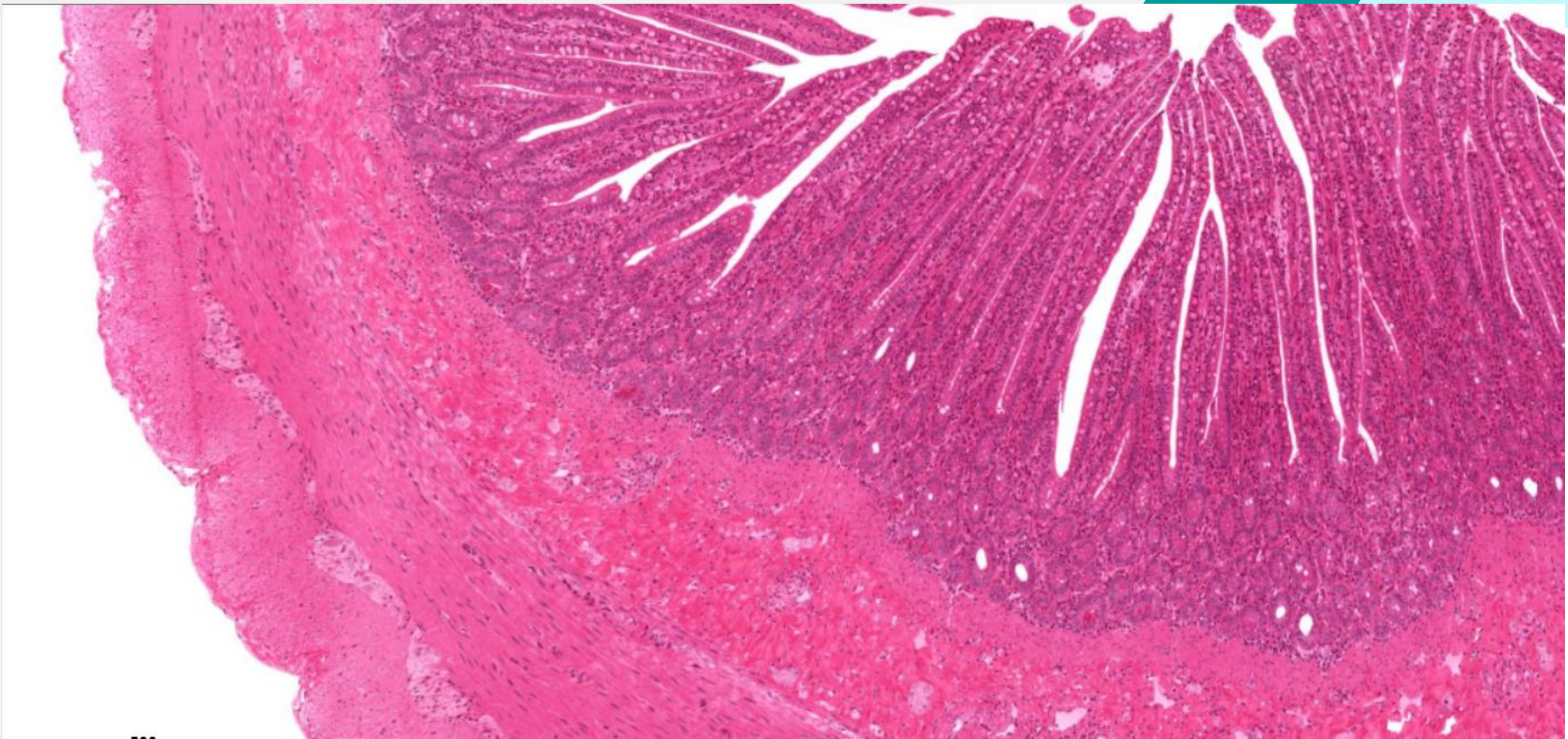
Regional differences between different parts of small intestine

	Duodenum	Jejunum	Ileum
Villi	Narrow, taller and more numerous than that of jejunum & ileum	Narrow, shorter and less than those of duodenum	Broad, shortest and least in number
Goblet cells	Has fewer goblet cells per unit area	Number of goblet cells is greater than in the duodenum	Goblet cells are numerous, more than those in duodenum & jejunum
Gland	Brunner's glands in submucosa	Absent	Absent
Lymphoid nodules	Absent	Absent	Clusters of lymphoid nodules (Peyer's patches) in lamina propria

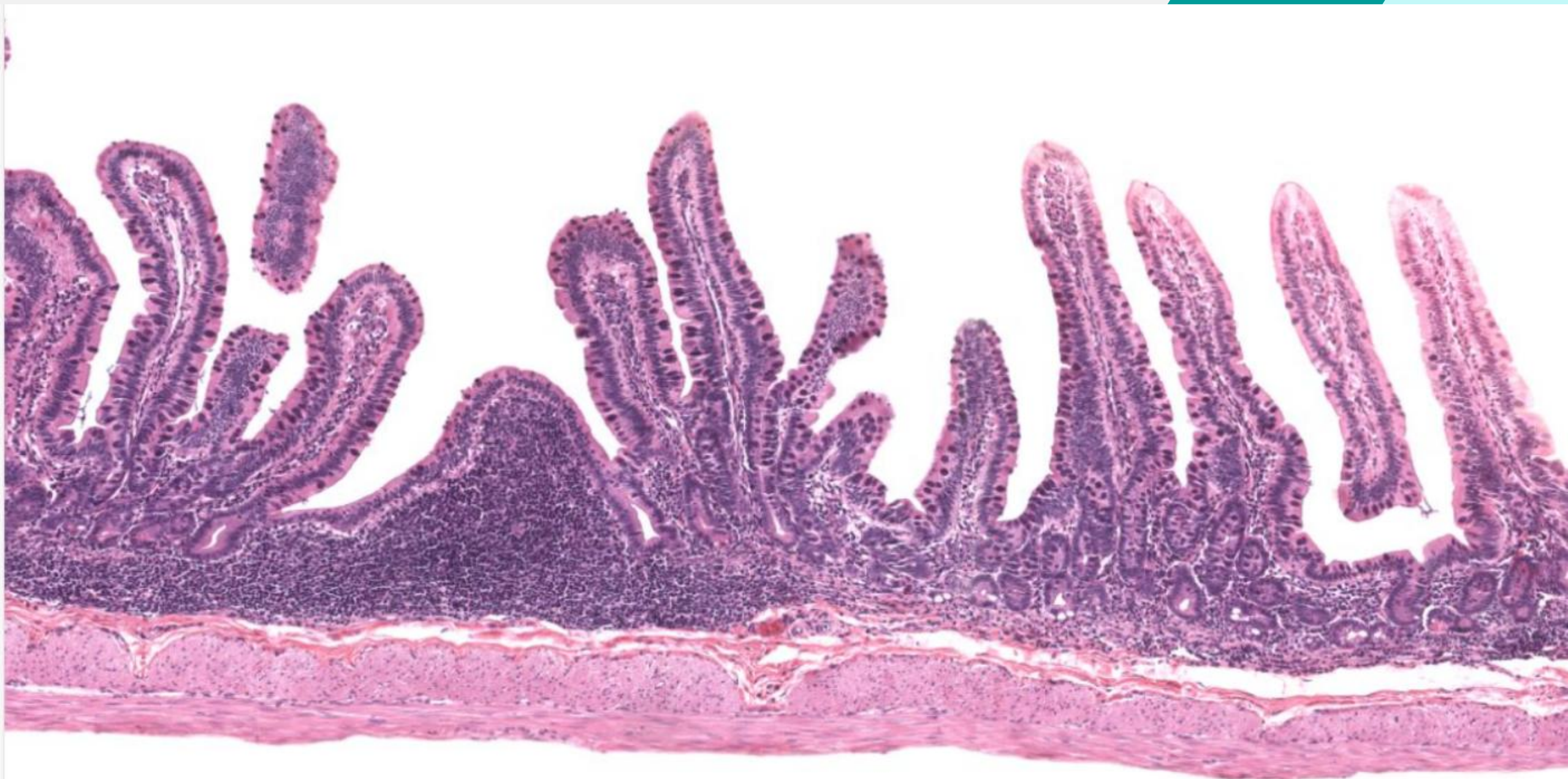
Duodenum

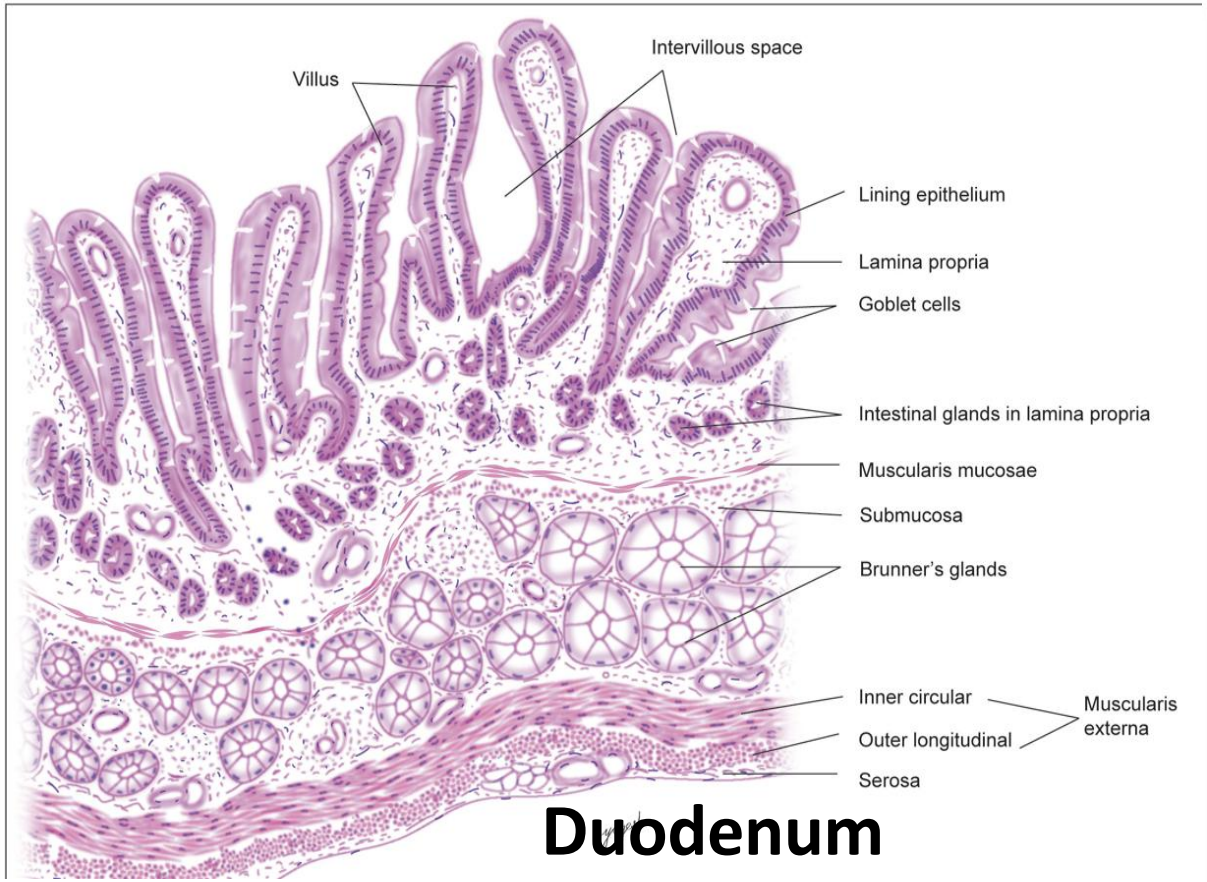


Jejunum

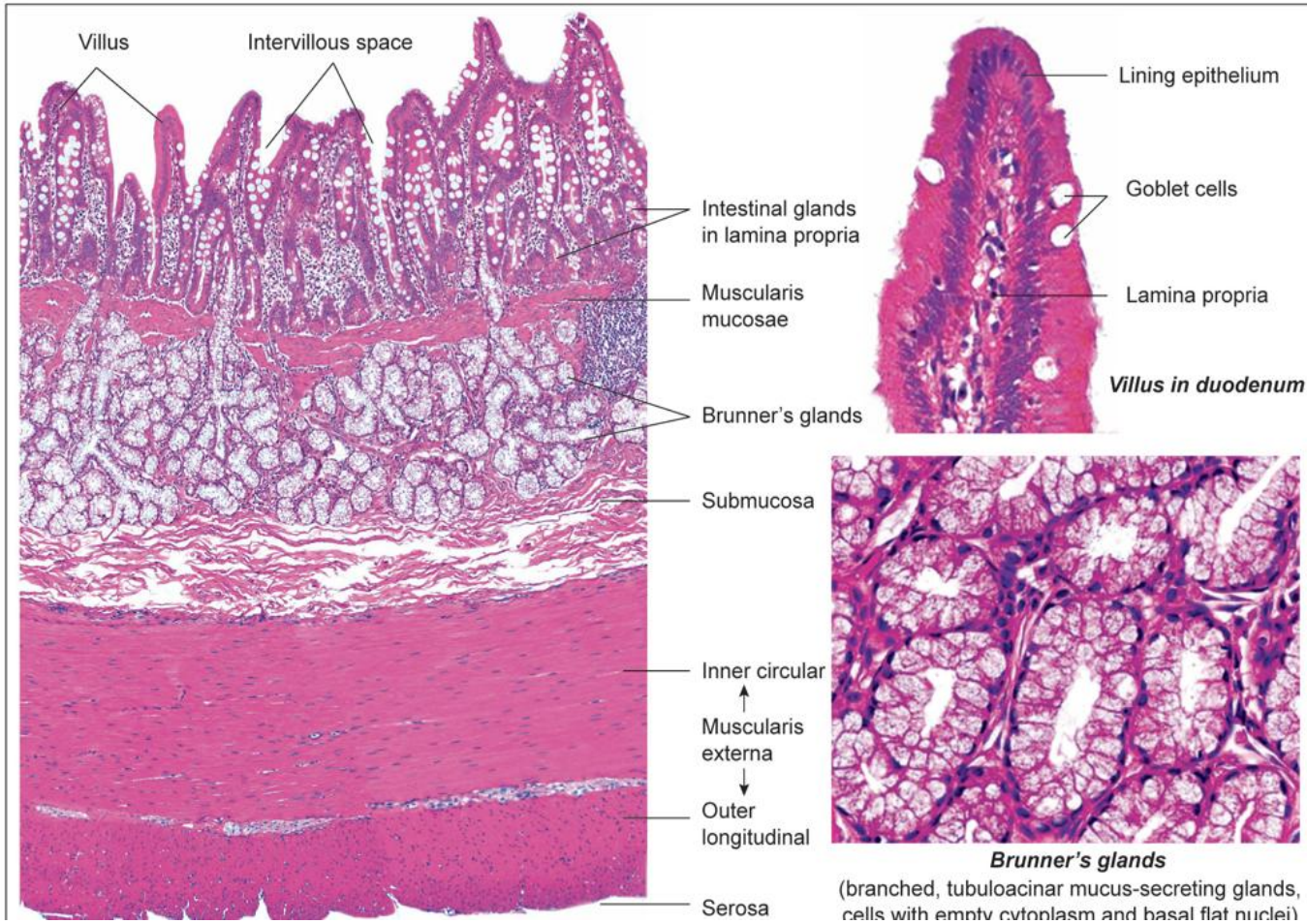


Ileum





Duodenum



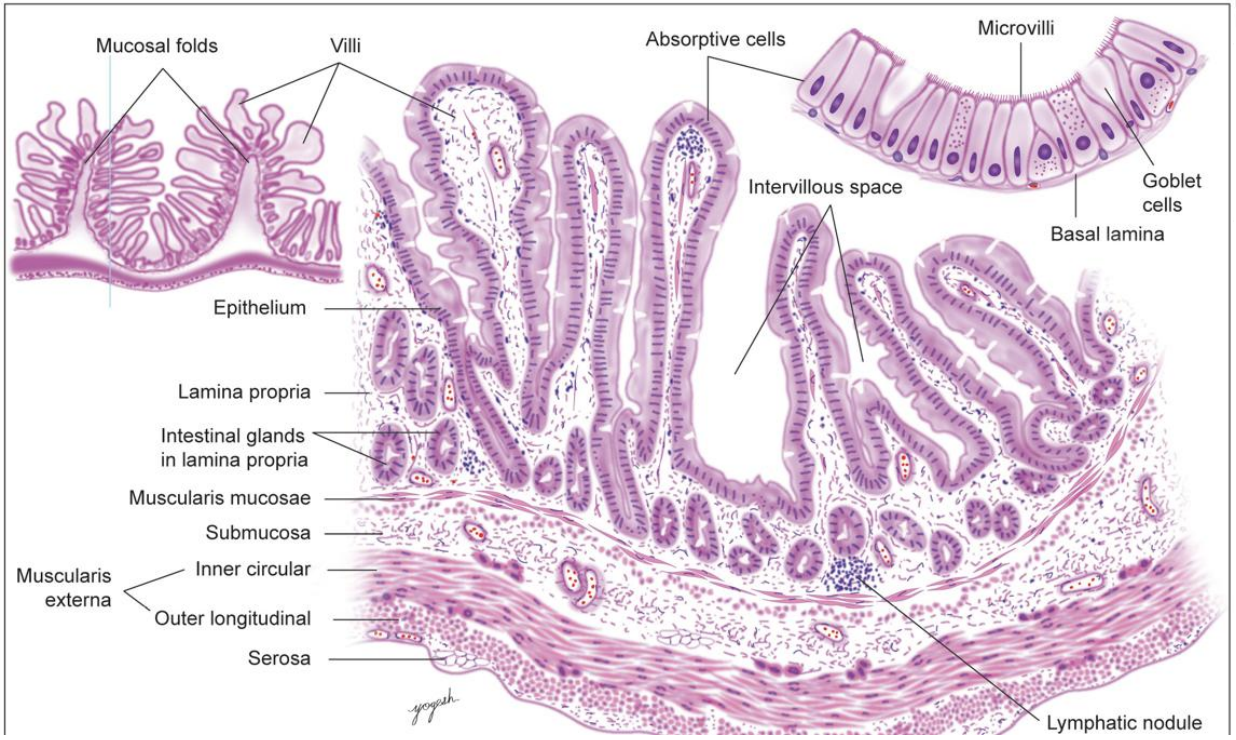
Mucosa
- Tall columnar epithelium with striated border and few goblet cells
- Finger-like villi
- Crypts of Lieberkühn in lamina propria
- Muscularis mucosae

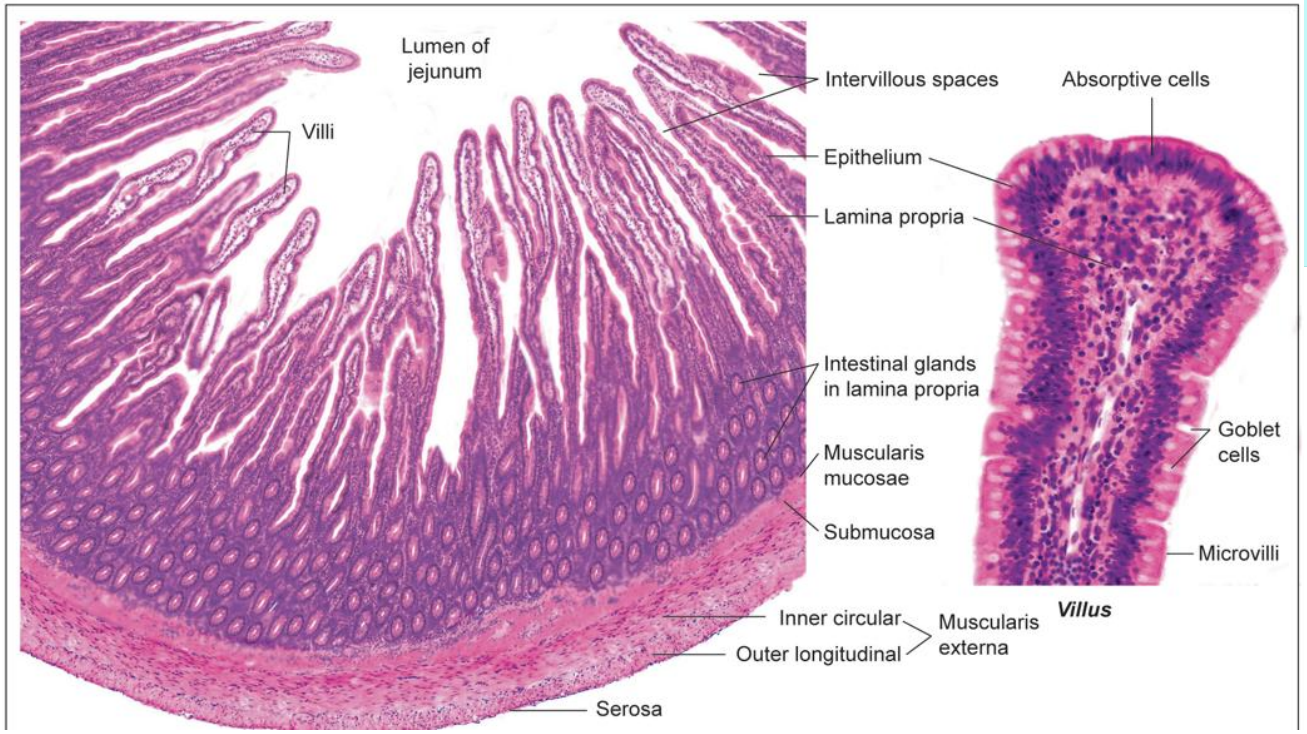
Submucosa
- Brunner's glands: Highly branched tubuloalveolar glands
- Plicae circularis: Circular folds of submucosa

Muscularis externa

Smooth muscle layer
- Inner circular
- Outer longitudinal

Adventitia/Serosa
- Connective tissue layer
- Proximal 1 inch and distal most part has serosa





Jejunum

Mucosa

- Tall columnar epithelium with striated border and few goblet cells
- Tongue-shaped villi
- Crypts of Leiberkuhn in lamina propria
- Muscularis mucosae

Submucosa

- Connective tissue
- No Brunner's glands and Peyer's patches

Muscularis externa

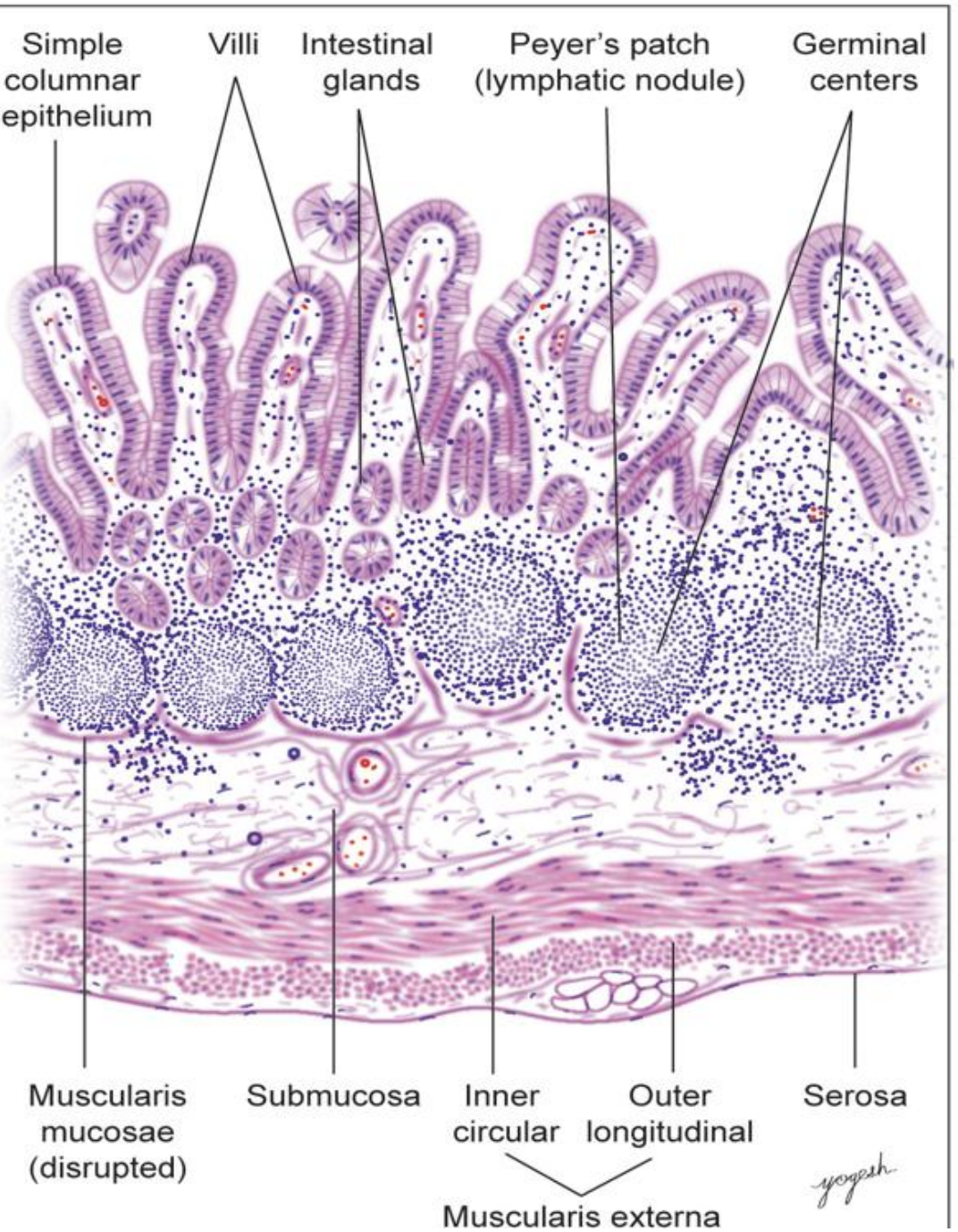
Smooth muscle layer
- Inner circular
- Outer longitudinal

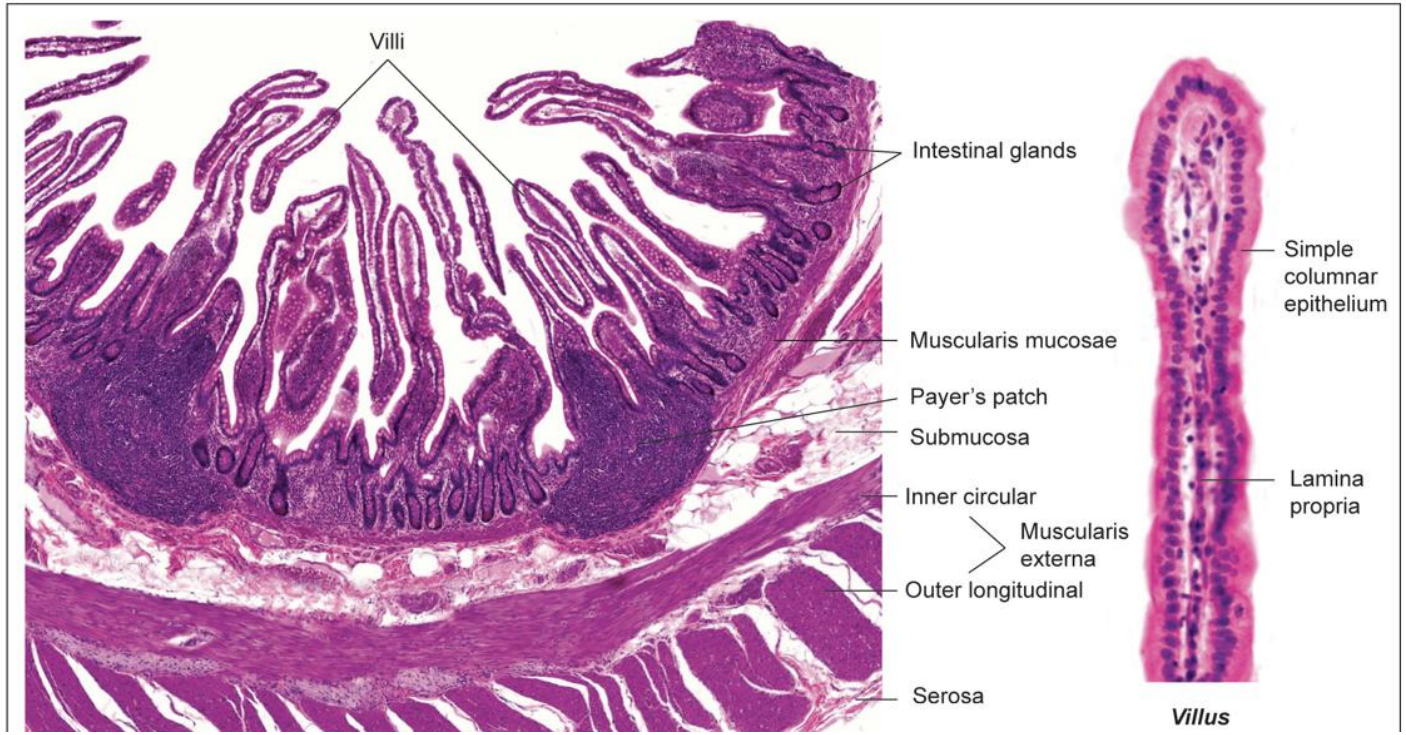
Serosa

Connective tissue layer
Covered by mesothelium

Note: Jejunum has larger diameter, thicker wall, and numerous large plicae circularis, and larger villi than ileum. *Identification feature*

Ileum


Simple columnar epithelium
Villi
Intestinal glands
Peyer's patch (lymphatic nodule)
Germinal centers
Muscularis mucosae (disrupted)
Submucosa
Inner circular
Outer longitudinal
Muscularis externa
Serosa
yogesh


Villi
Intestinal glands
Muscularis mucosae
Payer's patch
Submucosa
Inner circular
Muscularis externa
Outer longitudinal
Serosa
Simple columnar epithelium
Lamina propria
Villus

LARGE INTESTINE

Colon is the terminal segment of large intestine, mostly retroperitoneal.

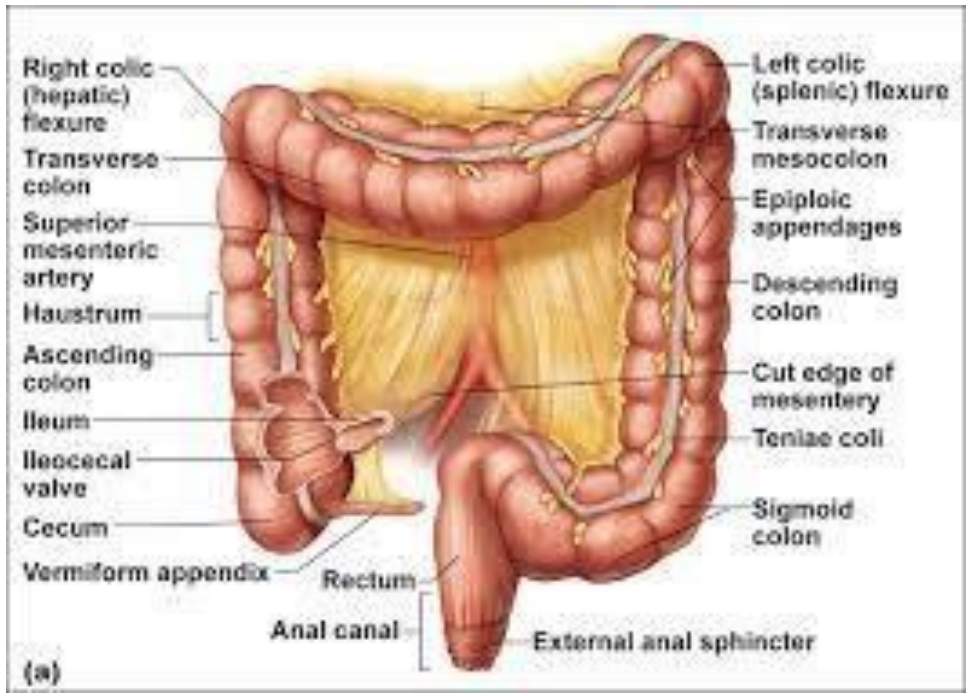
Functions:

✓Absoroption of water

✓Conversion of liquid,undigested material into solid faeces.

The diametre of large intestine is more than small intestine.

External features



MICROSCOPIC FEATURES



Mucosa:
Folded in non distended state. No villi and plicacircularis

Lining epithelium

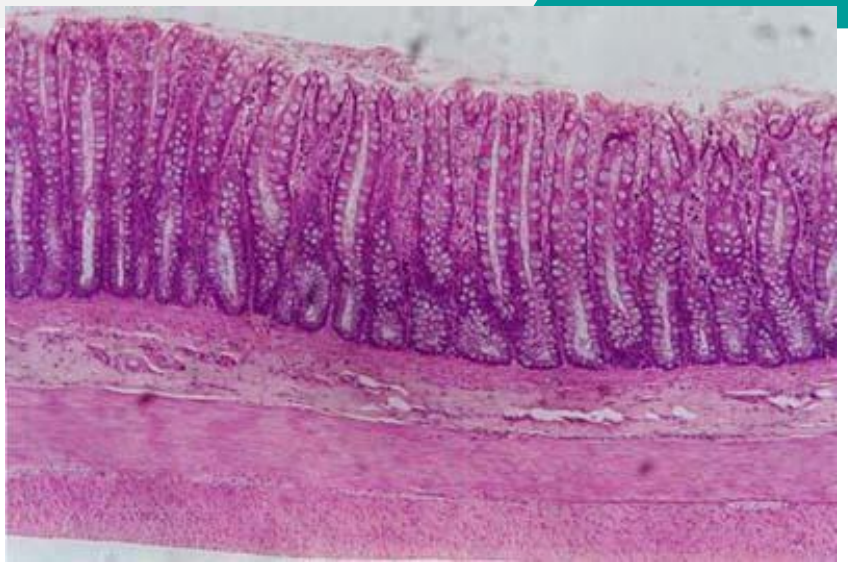
Simple columnar epithelium with more goblet cells

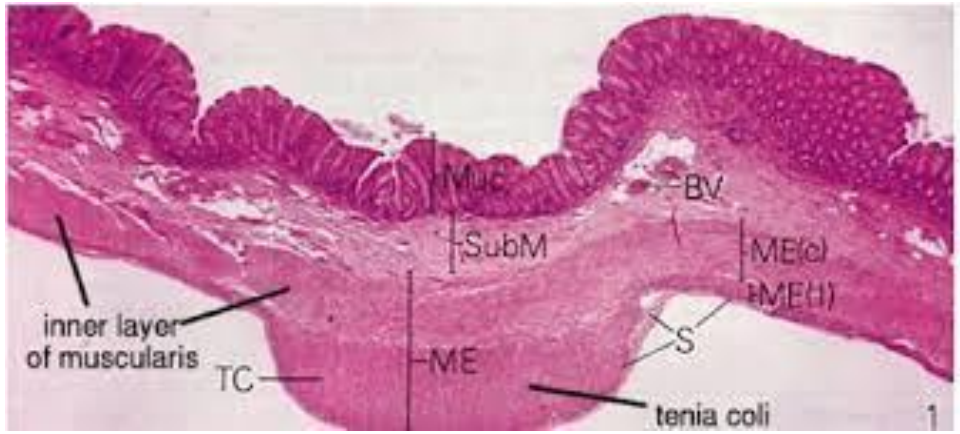
Microvilli is seen in absorptive cells.

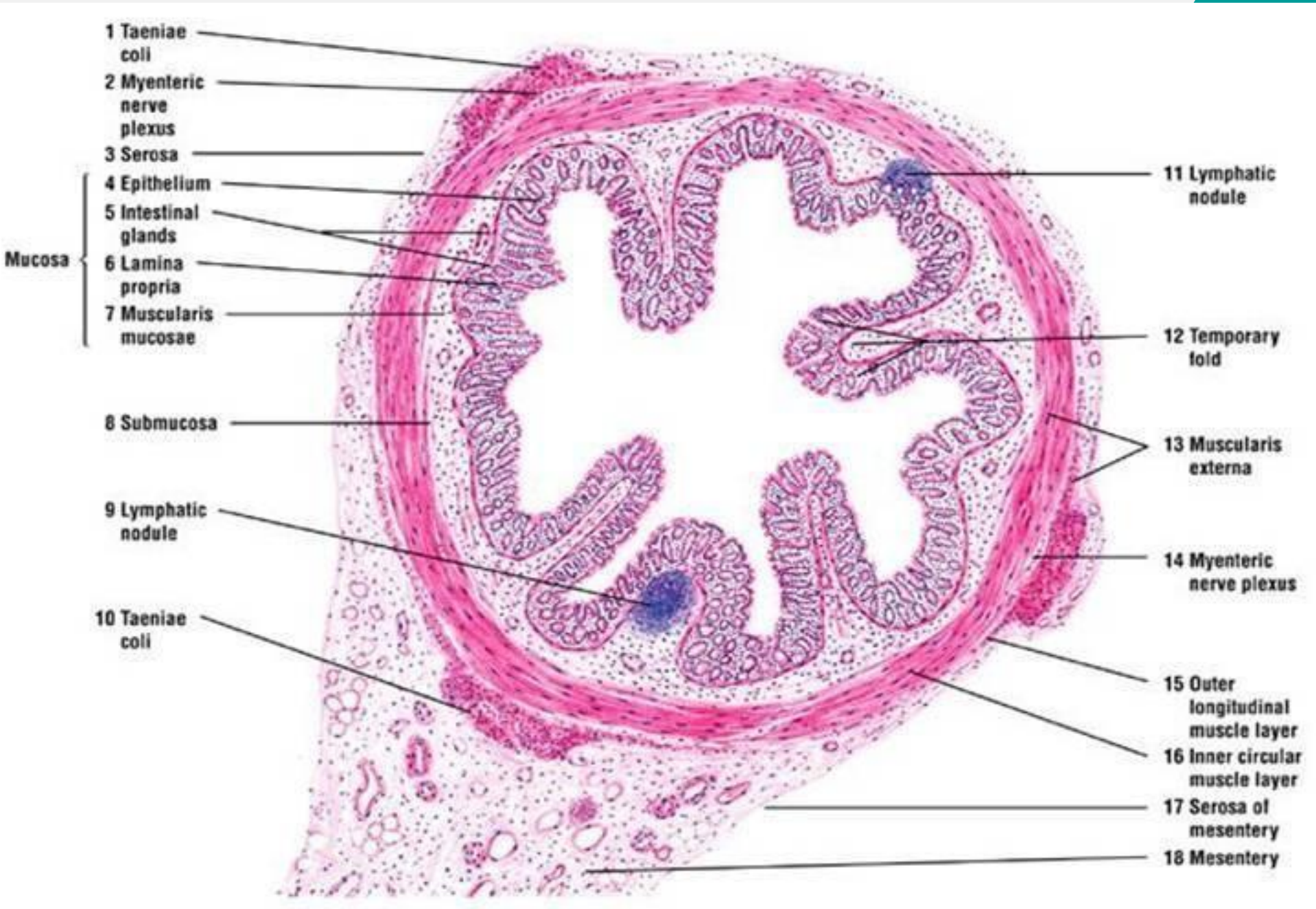
Contains straight unbranched tubular glands-crypts of lieberkuhn Enteroendocrine & stem cells seen in lower part of gland (no paneth cells)

Lamina propria
Rich in lymphocytes(GALT)
Muscularis mucusa

Submucosa:
Dense iregular connecive tissue.
Large blood vessels, lymphatics
adipose tissue and *meisner's plexus*.
Muscularis externa:
Inner circular and outer longitudinal
layer of smooth muscle
Taenia coli
Serosa/adventitia






1 Taeniae coli
2 Myenteric nerve plexus
3 Serosa
Mucosa
4 Epithelium
5 Intestinal glands
6 Lamina propria
7 Muscularis mucosae
8 Submucosa
9 Lymphatic nodule
10 Taeniae coli
11 Lymphatic nodule
12 Temporary fold
13 Muscularis externa
14 Myenteric nerve plexus
15 Outer longitudinal muscle layer
16 Inner circular muscle layer
17 Serosa of mesentery
18 Mesentery

APPENDIX

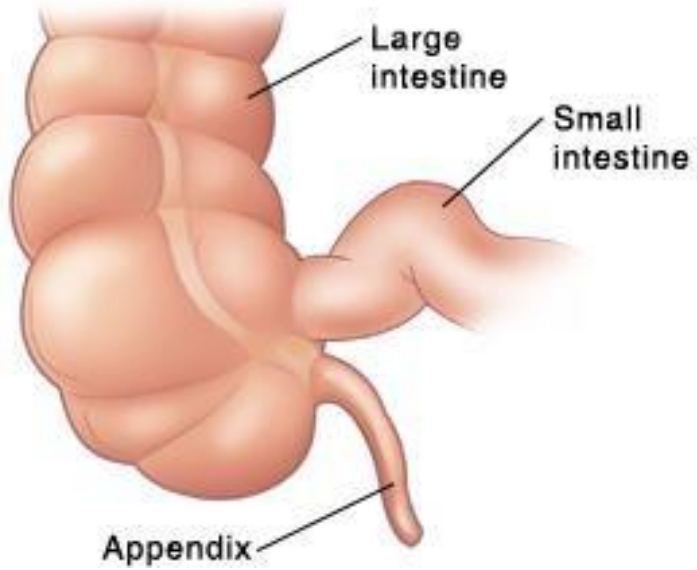
Worm like tubular diverticulum .

Posteromedial wall of caecum.

It’s the narrowest part of the gut.

Although it’s a part of colon, it is devoid of taenia coli ,sacculations & appendices epiploicae

Abdominal tonsil



MICROSCOPIC FEATURES

Resembles colon

Mucosa

Straight tubular glands(with goblet cells) which extends as far as muscularis.

But they are fewer and shorter.

Crypts are poorly formed.

Lamina propria-lymphocytes

Muscularis mucosae-obscure

Submucosa

Large aggregations of lymphocytes.

Muscularis externa

Serosa

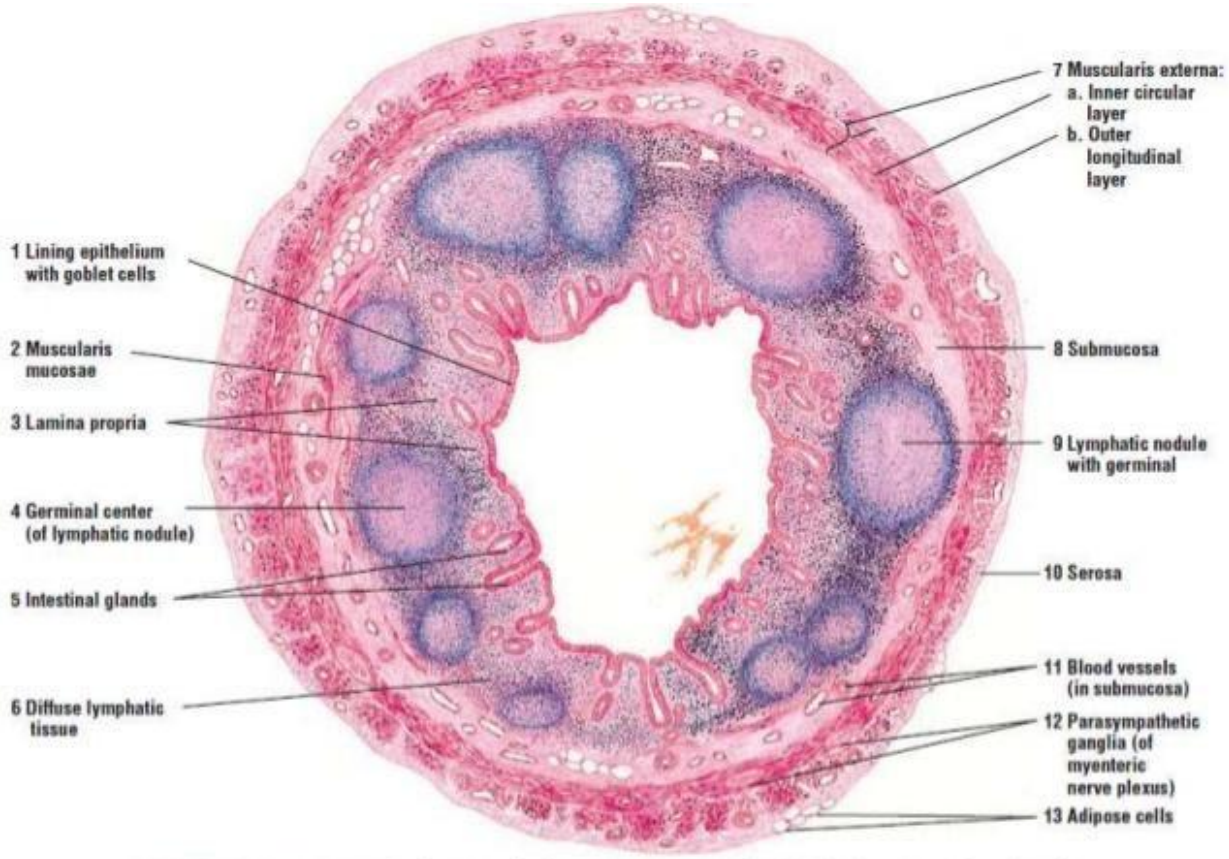


FIGURE 13.11 ■ Appendix (panoramic view, transverse section). Stain: hematoxylin and eosin. Low magnification.

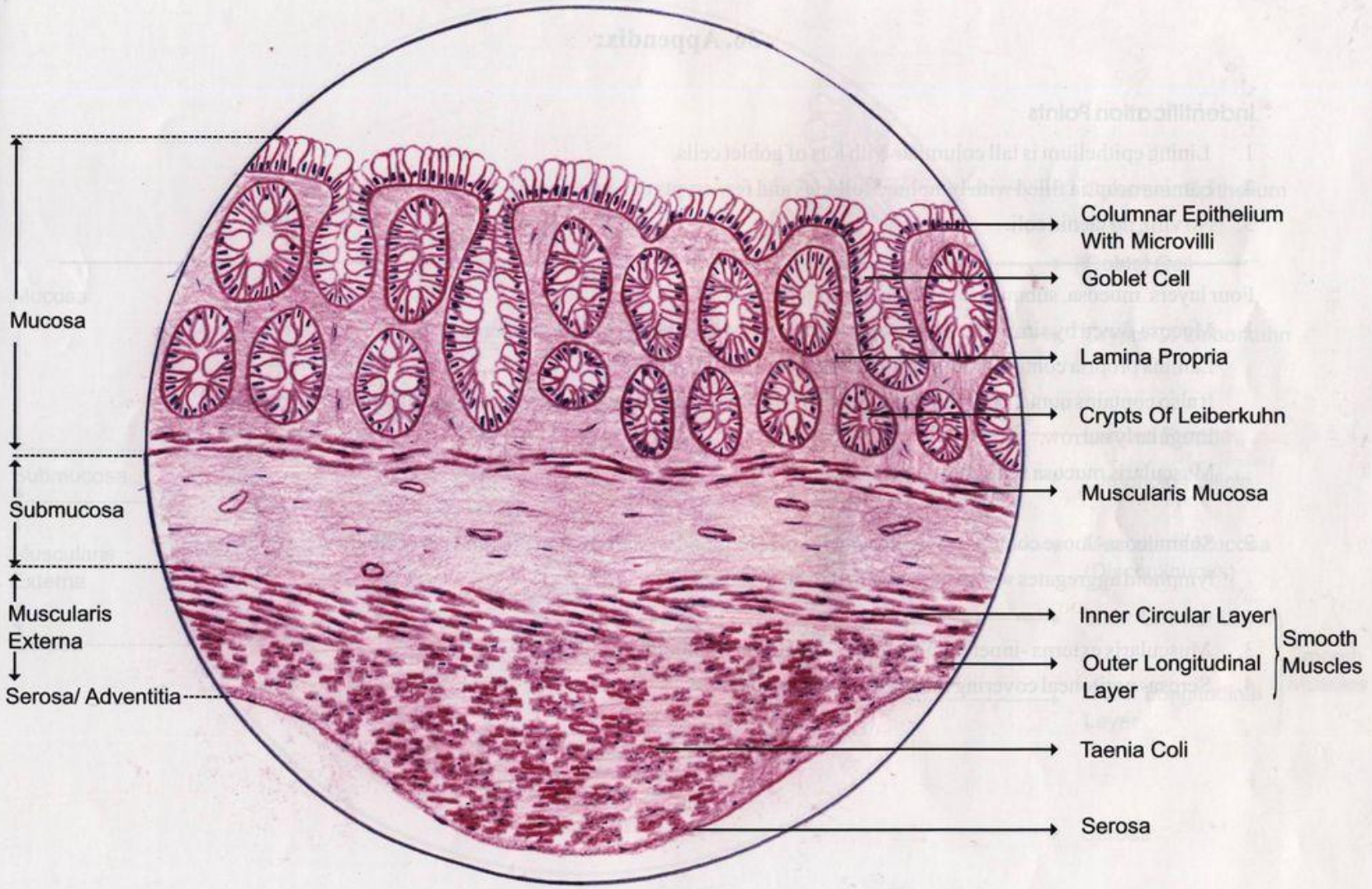
Large Intestine

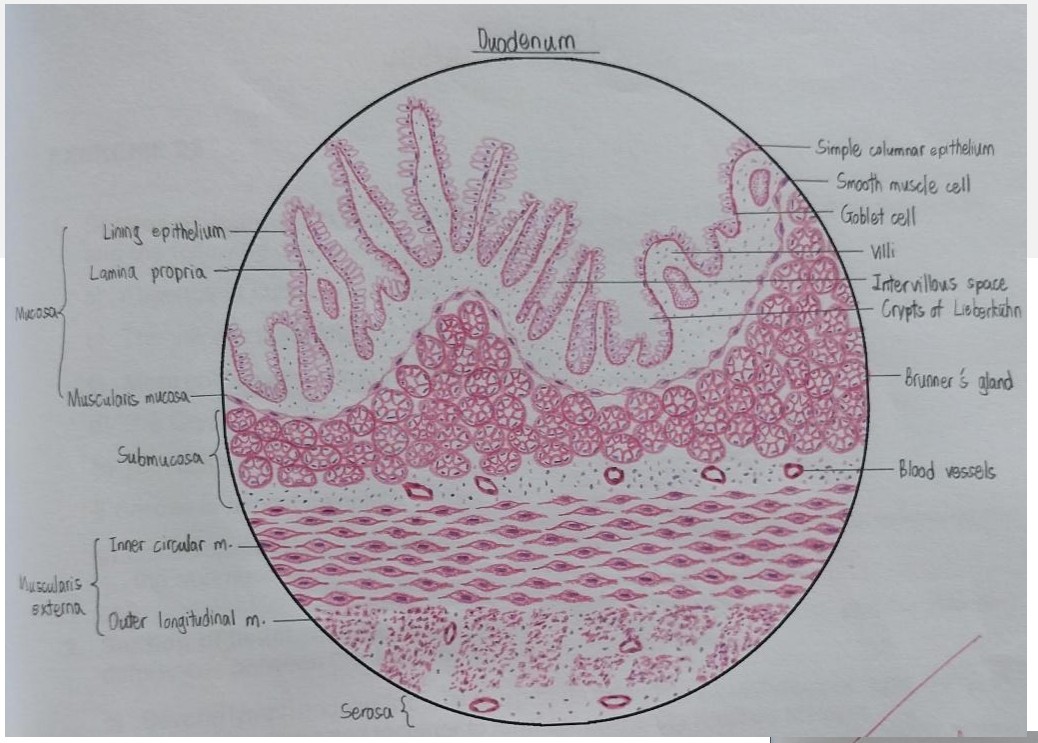


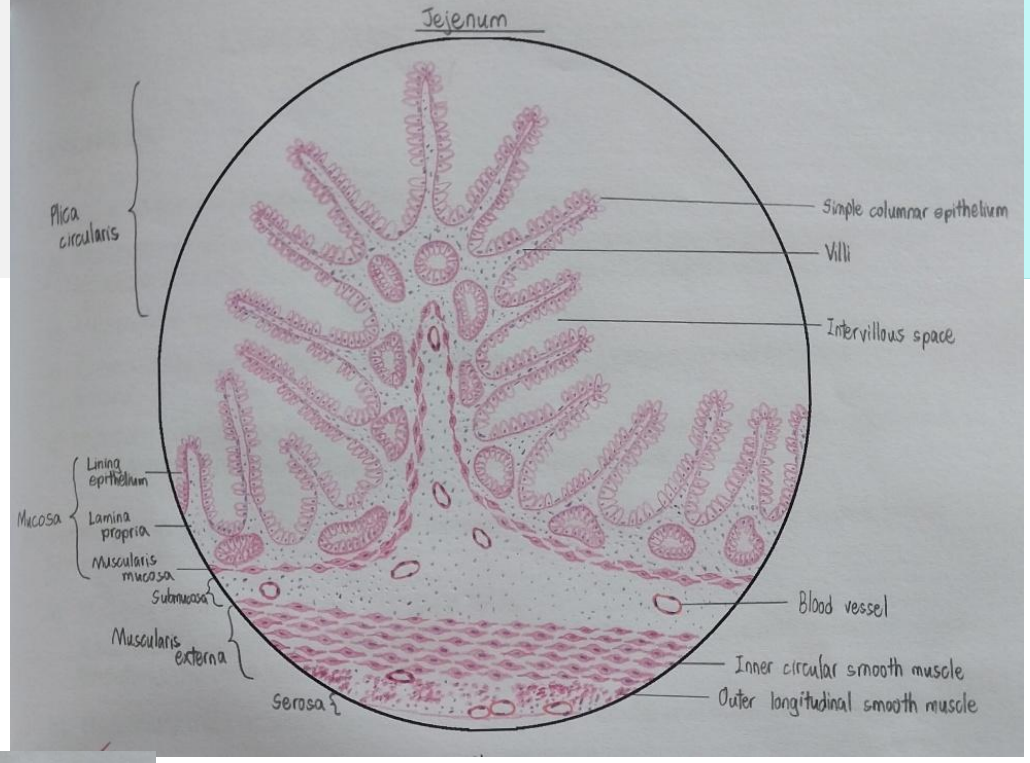
Appendix

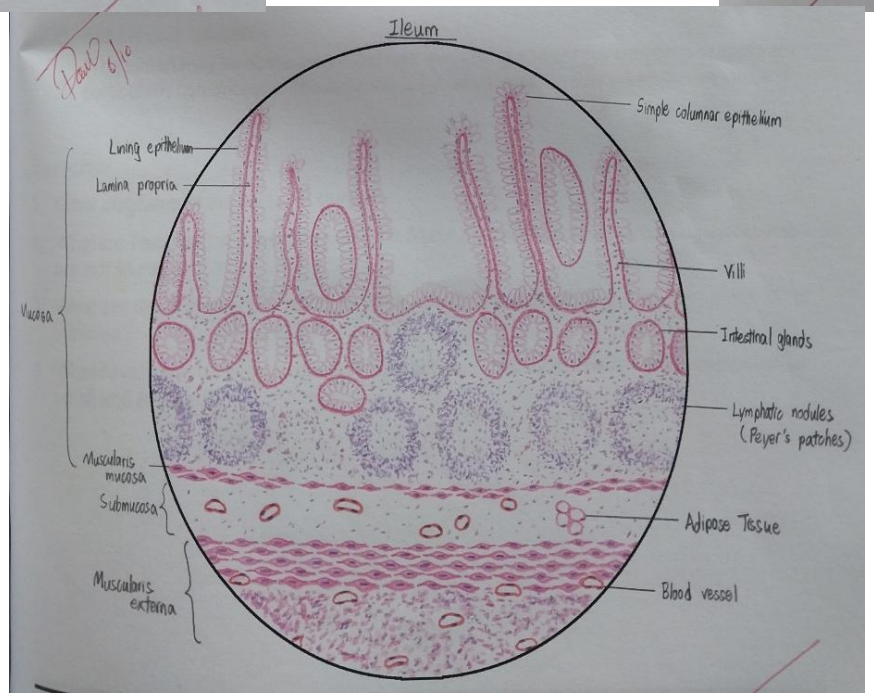


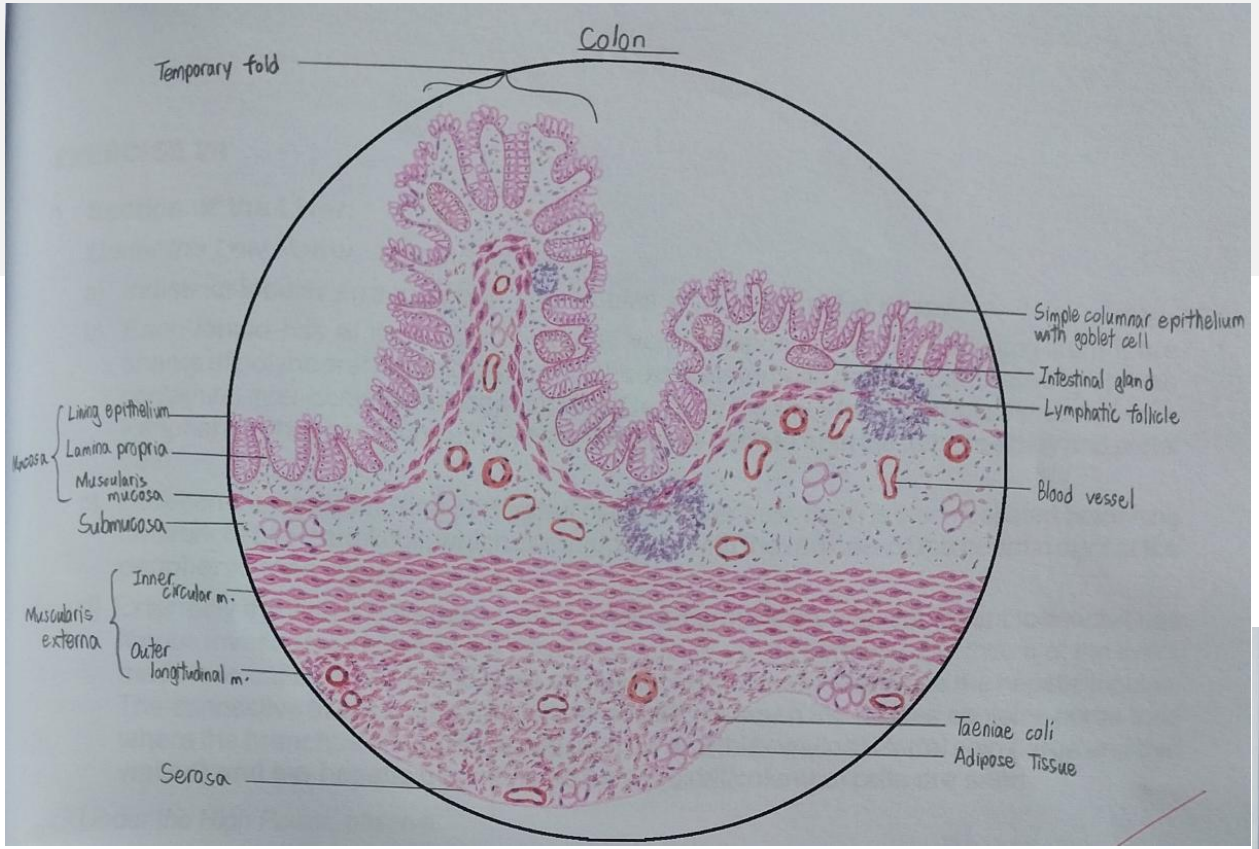
FIG NO. 35 LARGE INTESTINE

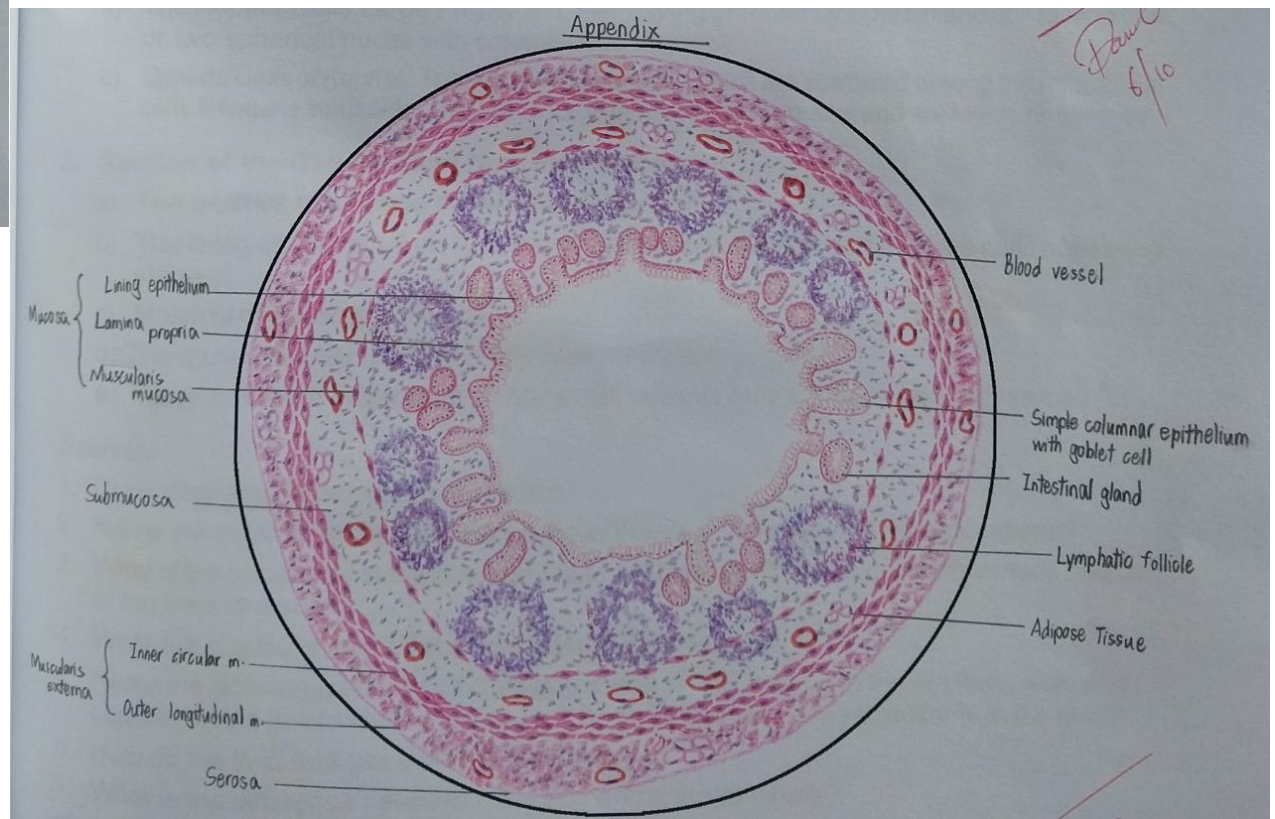



Duodenum
Simple columnar epithelium
Smooth muscle cell
Goblet cell
Villi
Intervillous space
Crypts of Lieberkühn
Brunner's gland
Blood vessels
Lining epithelium
Lamina propria
Mucosa
Muscularis mucosa
Submucosa
Inner circular m.
Outer longitudinal m.
Muscularis externa
Serosa


Jejenum
Plica circularis
Simple columnar epithelium
Villi
Intervillous space
Lining epithelium
Lamina propria
Mucosa
Muscularis mucosa
Submucosa
Blood vessel
Muscularis externa
Inner circular smooth muscle
Outer longitudinal smooth muscle
Serosa


Ileum
Simple columnar epithelium
Lining epithelium
Lamina propria
Villi
Mucosa
Intestinal glands
Lymphatic nodules (Peyer's patches)
Muscularis mucosa
Submucosa
Adipose Tissue
Muscularis externa
Blood vessel


Colon
Temporary fold
Simple columnar epithelium with goblet cell
Intestinal gland
Lymphatic follicle
Blood vessel
Mucosa
Lining epithelium
Lamina propria
Muscularis mucosa
Submucosa
Muscularis externa
Inner circular m.
Outer longitudinal m.
Taeniae coli
Adipose Tissue
Serosa


Appendix
Blood vessel
Mucosa
Lining epithelium
Lamina propria
Muscularis mucosa
Simple columnar epithelium with goblet cell
Intestinal gland
Submucosa
Lymphatic follicle
Adipose Tissue
Muscularis externa
Inner circular m.
Outer longitudinal m.
Serosa